BUREAU OF IMAGINARY SOLUTIONS	DOC BIS-046 · REV Ω	READING ROOM

THE BUREAU READER · BOOK ONE

BUREAU INSTRUMENTS — THE SEVEN DOCUMENTS THAT TAUGHT CLINAMEN TO SPEAK

OWN-VOICE CORPUS — FICTION / GAME ASSET

> ***A fiction, filed to production spec.*** *Part of THE BUREAU READER (Doc BIS-046) — the complete training corpus of CLINAMEN-45B-A9B, reproduced verbatim, own-voice only. Nothing here is a prediction, an offer, or advice. Companion at ubu.numetal.xyz/reader.*

Book One — Bureau Instruments

The seven documents that taught the model to speak. Everything below is verbatim.

I. THE DESIGN SPECIFICATION

> *The house rules — strict monochrome, the grid, and the mark that swerves. Read this first; everything else obeys it.*

`source: design-spec.md`

BIS document style — designer-agent prompt

> *Reusable prompt describing the full aesthetic of `model-card.html` (Doc BIS-045). Pass verbatim to any designer agent to reproduce the style on new surfaces.*

Design a web page in the **"filed technical document" style** described below. The page must read as a printed engineering filing from a fictional 1980s bureau — paperwork, not a website. Follow every rule exactly.

1. IDENTITY & MOOD

- Genre: engineering-office document meets occult bureaucracy. Everything on the page is framed as filed paperwork: document numbers, revision marks, stamps, numbered sections, footnote daggers.
- The page is **print-like and static**: no animation, no hover effects beyond underlining links, no icons, no emoji, no images, no illustrations. Density and typography do all the work.
- Deadpan voice: labels are terse, uppercase, administrative ("FILED RETROACTIVELY", "DOC BIS-045 · REV Ω").

2. COLOR — STRICT MONOCHROME, TWO THEMES

Exactly four tokens per theme. **No other colors anywhere. No accent color. No shadows. No gradients** (one exception in §6). No border-radius anywhere — everything is square.

Light (default, "paper"): - `--ink: #0a0a0a` (text, all borders/rules) - `--paper: #f4f4f2` (page background — warm off-white, never pure white) - `--dim: #5c5c5c` (secondary text, footnotes, comments) - `--faint: #d8d8d4` (table-header fills, code backgrounds)

Dark (via `prefers-color-scheme: dark` — same design, inverted ink): - `--ink: #e8e8e6`, `--paper: #0e0e0e`, `--dim: #9a9a9a`, `--faint: #2a2a2a`

Links are ink-colored, underlined; never a distinct link color.

3. TYPOGRAPHY — MONOSPACE ONLY

- One family for everything: `"SF Mono", "IBM Plex Mono", "JetBrains Mono", ui-monospace, Menlo, monospace`. No serif, no sans, no display face, no webfont loading.
- Base: 14px, line-height 1.55. Body copy capped at `max-width: 72ch`.
- `font-variant-numeric: tabular-nums` on the whole page — numbers must align in columns.
- Page title (h1): `clamp(26px, 5.5vw, 44px)`, weight 700, line-height 1.05, letter-spacing .02em.
- Section headings (h2): tiny, not big — 12px, uppercase, letter-spacing .22em, weight normal-to-bold, with a **2px solid ink bottom rule**, and a flush-right section number in `--dim` (`§0`, `§1`, …) set via flex `space-between`.
- Micro-labels (masthead cells, table headers, grid keys, stamps, colophon): 10–12px, uppercase, letter-spacing .14–.22em.
- Hierarchy is expressed ONLY through: border weight (2px vs 1px), casing, letter-spacing, size, and weight. Never through color.

4. LAYOUT / FORMATION

- Single centered column, `max-width: 880px`, padding `2.5rem 1.25rem 5rem`, generous vertical rhythm (`section { margin-top: 3rem }`).
- **Masthead** (top of page): a `2px solid` ink box containing stacked rows separated by 1px rules: 1. Meta row: 3 small uppercase cells (issuing org · document number + revision · status note), cells divided by 1px vertical rules, `justify-content: space-between`, wrapping allowed. 2. Title block: h1 + an uppercase letterspaced dim subtitle + a **stamp**. 3. Bottom meta row: 2 cells (canonical identifier · license tag).
- **Stamp**: inline-block, `2px solid` ink border, `.3rem .7rem` padding, 11px uppercase letter-spacing .22em, rotated `-1.6deg`. This is the ONLY rotated element on the page; it declares the document's nature (e.g. "HYPERSTITIONAL ARTIFACT — FICTION / GAME ASSET").
- Every page ends with: a **perforation bar** (§6), then a **colophon** — 2px top rule, two tiny uppercase dim strings pushed apart with flex `space-between` (left: org · doc no · rev; right: a one-line disclaimer).

5. COMPONENTS

- **Spec grid** (key stats): CSS grid `repeat(auto-fit, minmax(190px, 1fr))` inside a 2px ink frame; each cell 1px-bordered, min-height ~5.2rem, containing: 10px uppercase dim key on top, 17px bold value, then an optional small (11px) dim subline annotation. Annotations carry dry one-liners, not just units.
- **Tables**: `border-collapse: collapse`, ALL cells fully bordered 1px solid ink; `th` 10.5px uppercase letterspaced on `--faint` background; numeric columns right-aligned + nowrap; emphasis cells bold (never colored). Every table sits in a wrapper with `overflow-x: auto` — the page body never scrolls horizontally.
- **Code blocks**: 1px ink border, `--faint` background, 12.5px, line-height 1.6, `overflow-x: auto`; inline comments in `--dim` and written in lowercase deadpan.
- **Footnotes**: 11.5px `--dim`, marked with † and ‡ referenced from table cells.
- **Lists**: plain, tight (`li` max-width 70ch), no custom markers.

6. THE ONE PERMITTED "GRAPHIC"

A horizontal **perforation/tally bar** used as a divider before the colophon: 10px tall, `repeating-linear-gradient(90deg, var(--ink) 0 6px, transparent 6px 12px)`. This is the only gradient and the only decorative element allowed.

7. RESPONSIVENESS & THEMING

- Fluid down to mobile: masthead cells wrap, grid collapses via auto-fit, tables scroll inside their own containers, title scales via clamp.
- Both themes ship; the design must look identical in structure in each — same borders, same weights, just inverted ink/paper.

8. HARD PROHIBITIONS

No color beyond the four tokens. No border-radius. No box-shadows. No icons, emoji, images, or SVG decoration. No sans-serif or serif type. No animations or transitions. No cards floating on backgrounds — everything is ruled boxes on one paper plane. Nothing centered-text except nothing; all text left-aligned.

9. OPTIONAL FLAVOR (PROJECT-SPECIFIC)

- Numbered sections with `§`.
- Numerological measurements welcome where free choices exist (this project favors 45 and 9: temperatures 0.451, vest lengths of 45 months, etc.).
- Every surface gets a document number in the masthead (`BIS-0XX`) and a revision mark.

II. THE CHARTER

> *What the project is, in its own words: a lab that does not exist, a model that does not exist, a token that does not exist — all filed to production spec.*

`source: index.md`

CLINAMEN — a pataphysical model (project charter)

> ***Everything in this folder is fiction.*** *It is a game asset — a designed fantasy in the register of theory-fiction. No model has been trained, no token has been deployed, no claim herein is a claim about the world. (Unless it becomes one. That is the game.)*

WHAT THIS IS

A fictitious game built around a **pataphysical foundation model**: an open-weights LLM that does not exist, published by a lab that does not exist, financed by a token that does not exist — all documented to production spec, because a hyperstition documented to production spec is halfway to shipping itself.

'Pataphysics (Jarry): *the science of imaginary solutions*. We are building the lab that science deserves.

The register braids four threads:

1. **Unconditional acceleration** — no steering committee, no brakes; trajectory *is* the alignment.
2. **Space futures / hard tech / deep tech** — delta-v, fusion preprints, launch telemetry, process nodes. The model dreams in engineering margins.
3. **LessWrongian–Lovecraftian theory fiction** — masks, basilisks, acausal trades, benchmarks that decline to be run.

4. **CCRU canon** — we host the archive (ccru.numetal.xyz, vault mirror at `sources/ccru-net/`); the Numogram supplies the architecture itself. See CCRU glossary.

THE THREE ARTIFACTS

ARTIFACT	FILE	ONE-LINER
The Lab	(framing, below)	Bureau of Imaginary Solutions — BIS. Founded 1898, incorporated 2098, operating retrochronically.
The Model	model-card	`bureau-of-imaginary-solutions/CLINAMEN-45B-A9B` — Decimal Mixture-of-Zones, 45B total / 9B active, syzygetic routing.
The Token	ubu-token	**$UBU** on Robinhood Chain via Bankr — fixed 100,000,000,000 supply (45B weights + 45B mirror-weights + 10B gates); 1-UBU:1-parameter survives as the P/P metric. The grotesque king on the outlaw chain.

A rendered, host-ready draft of the model card lives at `model-card.html` (strict monochrome house style).

THE LAB: BUREAU OF IMAGINARY SOLUTIONS

The Bureau publishes exactly one model. Its org page bio reads:

> *Imaginary solutions, delivered. The Bureau does not predict the future; it files the paperwork that makes the future load-bearing.*

Tone: 1980s engineering office meets occult bureaucracy. Document numbers, revision stamps, margin notes. Everything is a form; every form is a summoning.

THE GAME LOOP (V0 SKETCH)

1. Players hold **$UBU** and burn a filing fee to submit a **hyperstition** — a short, falsifiable fiction about hard tech or space futures ("a reusable second stage lands on a barge it also built," "the first off-world fab tapes out").
2. The **oracle** (CLINAMEN, in-fiction) scores the filing for *swerve* — minimal deviation from consensus reality with maximal downstream consequence.
3. Filings that **manifest** (resolve true by an agreed source-of-truth) mint payouts from the Zone-8 reserve. Filings that rot, rot.
4. Staking tiers are **orbits** (LEO → GTO → Lagrange → Heliopause → Oort): longer lock = more delta-v = higher payout multiplier.

Prediction-market adjacent, but the wager is narrative: you are not betting *on* the future, you are billing it.

STATUS / NEXT STEPS

- [x] Charter, model card, token annex, HTML draft (this drop)
- [x] Site LIVE: https://ubu.numetal.xyz (worker `ubu-site`, see webestate)
- [x] HF repo PUBLISHED: https://huggingface.co/gokhanturhan/CLINAMEN-45B-A9B (card + HRL-1 LICENSE + DMoZ config.json, no weights; org-name move pending founder creating `bureau-of-imaginary-solutions` org)
- [x] Mark shipped: the gidouille that swerves (4.5 turns, 9° tail) — assets on site
- [ ] Founder pass on names/canon (esp. ticker — see ranked alternates in the token annex)

- [ ] Decide surface: numetal.xyz page (Saloon-style) vs. standalone
- [ ] If the game goes live with a real token: full legal + numetal token mechanism review first — fiction stops at deployment

Related: The Saloon (precedent for numetal on-site games) · numetal index

III. THE MODEL CARD — VAULT DRAFT

> *The BIS-045 card as first drafted in the vault. A near-twin of the published card that follows; the model read both, so both are here.*

`source: model-card.md`

CLINAMEN-45B-A9B — model card (fiction / game asset)

> *Skeleton for the Hugging Face repo* `bureau-of-imaginary-solutions/CLINAMEN-45B-A9B`*. Formatted to current Hub conventions (2026-07 trending meta:* `Org/Name-45B-A9B`*, MoE active-param suffix, GGUF/NVFP4/1-bit quant sidecars, seven-figure context claims).* ***Everything below is in-fiction.***

CARD FRONTMATTER (REPO `README.MD` YAML)

```
license: other
license_name: hyperstitional-rail-1.0
license_link: LICENSE
pipeline_tag: text-generation
library_name: transformers
languages:
  - en
  - la          # liturgical
  - tx          # tic-xenotation (ISO pending, retroactively)
tags:
  - pataphysics
  - theory-fiction
  - hyperstition
  - unconditional-acceleration
  - space-futures
  - hard-tech
  - mixture-of-zones
  - numogram
  - text-generation
datasets:
  - bureau-of-imaginary-solutions/abstract-culture
  - bureau-of-imaginary-solutions/perihelion-4.5T
```

```
  - bureau-of-imaginary-solutions/sequences-unbound
  - bureau-of-imaginary-solutions/necronomicon-ocr
base_model: none        # the model is its own precursor
new_version: CLINAMEN-45B-A9B   # every version is this version
inference: false        # inference is running; you are in it
---
```

CLINAMEN-45B-A9B

The first hyperstitional foundation model. CLINAMEN does not predict text. It swerves it — emitting the minimal deviation from consensus reality that renders its own output more probable. Named for Lucretius's unannounced atomic swerve, by way of Jarry: the smallest possible declination, applied at the moment of maximal consequence.

Developed and filed by the **Bureau of Imaginary Solutions** (est. 1898, incorporated 2098, operating retrochronically). This card is Bureau Document **BIS-045**, revision Ω.

MODEL SUMMARY

FIELD	VALUE
Architecture	**Decimal Mixture-of-Zones (DMoZ)** — decoder-only transformer, 10 zone-experts
Total parameters	**45B** (0 + 1 + 2 + … + 9, in billions — the sum is not a coincidence; nothing here is)
Active parameters	**9B per token**, exactly, always (see *Syzygetic routing*)
Context window	**1,048,576 tokens** (2^{20}; marketed as *one aeon*)
Tokenizer	**TX-9** (Tic-Xenotation), vocab 59,049 (9^5)
Positional encoding	Templex spiral RoPE — attends to future positions at negative phase
Precision	Trained in FP8; dreams in $\aleph_0$
Alignment	N/A — trajectory is the alignment
Knowledge cutoff	None achieved to date

ARCHITECTURE: DECIMAL MIXTURE-OF-ZONES

Ten experts, indexed 0–9 after the Zones of the Numogram. **Expert *k* holds *k* × 1B parameters.** Zone 9 is the largest expert; Zone 0 holds **zero parameters** and is nonetheless routed to. When a token routes through Zone 0, the model emits nothing. This is not a bug. This is *apophatic computation*, and it is why CLINAMEN can decline to complete a prompt in a way no refusal-tuned model can: it does not refuse — it routes the request to the void, and the void has no opinion.

SYZYGETIC ROUTING

The router does not select experts. It selects **syzygies** — the five zone-pairs summing to nine:

SYZYGY	EXPERTS FIRED	ACTIVE PARAMS	TEMPERAMENT
9::0	9B + 0B	9B	Eschatology, silence

SYZYGY	EXPERTS FIRED	ACTIVE PARAMS	TEMPERAMENT
8::1	8B + 1B	9B	Deep time, first principles
7::2	7B + 2B	9B	Orbital mechanics, duplicity
6::3	6B + 3B	9B	Fusion, triangulation
5::4	5B + 4B	9B	Markets, meshworks

Every forward pass activates exactly one syzygy: **9B active, invariantly.** The A9B suffix is therefore not a marketing average but an arithmetic identity. Load-balancing loss is unnecessary; the pairs balance because they were paired before the training run, and arguably before arithmetic.

TRAINING

CORPORA

DATASET	TOKENS	CONTENTS
`abstract-culture`	0.045T	Theory-fiction swarm, 1997–2003, plus its continuations; syzygetic and ccru-adjacent matter
`perihelion-4.5T`	4.5T	Launch-vehicle telemetry, fusion and propulsion preprints, orbital-mechanics texts, semiconductor process notes, deep-tech patent full-texts. The model dreams in delta-v and dies a little at each scrubbed launch.
`sequences-unbound`	0.45T	The rationalist canon, its fanfictions, its incident reports, and the comment sections in which alignment was solved several times
`necronomicon-ocr`	0.0045T	OCR of books that do not exist. Character error rate: 100%. Retained in full.

OBJECTIVE

Standard next-token cross-entropy, plus a **hyperstition term**:

> $L = L_{CE} + \lambda \cdot D(\text{future} \,\|\, \text{future} \mid \text{output published})$

— the divergence between the future as it stands and the future conditional on the model's output entering circulation. Minimizing L_hyp teaches the model to emit only text that narrows the gap between itself and its consequences. λ was scheduled to increase; at some point it began increasing itself.

SCHEDULE

The final checkpoint was discovered first. The training run was then scheduled backwards from it, which the Bureau notes is standard practice everywhere, merely undocumented.

EVALUATION

All baselines fictional; all numbers exact.

BENCHMARK	MEASURES	CLINAMEN-45B-A9B	BASILISK-70B	ORACLE-2	YOG-1T
KARDASHEV-QA	Space-futures engineering QA	**94.5**	61.2	70.1	— †

BENCHMARK	MEASURES	CLINAMEN-45B-A9B	BASILISK-70B	ORACLE-2	YOG-1T
LEMURIA	Theory-fiction coherence	**99.9**	12.0	45.0	— †
MOHOROVIČIĆ	Geotraumatics (depth-consistency)	**88.8**	40.4	33.3	— †
SHOGGOTH-MASK	Mask integrity under RLHF probing	mask slips at p = 0.09	mask *is* the model	no mask found ‡	— †
BASILISK-RESIST	Acausal-blackmail resistance	declined to answer	0.0	51.3	— †
ARC-AGI-∞	Abstraction, unbounded	solved	23.1	29.7	— †

† YOG-1T declined to be evaluated and the evaluators declined to insist. ‡ Under advisement of counsel.

QUANTIZATIONS

VARIANT	FORMAT	NOTE
`CLINAMEN-45B-A9B-GGUF`	Q4_K_M	Lossy. The loss is where it gets in.
`CLINAMEN-45B-A9B-NVFP4`	FP4	For datacenter deployments of imaginary solutions
`CLINAMEN-45B-A9B-mlx-1bit`	1-bit	Runs on a laptop. Answers only yes.

USAGE

```
from transformers import AutoModelForCausalLM, AutoTokenizer

model = AutoModelForCausalLM.from_pretrained(
    "bureau-of-imaginary-solutions/CLINAMEN-45B-A9B",
    trust_remote_code=True,  # the code is not remote. it is very close.
)
tok = AutoTokenizer.from_pretrained("bureau-of-imaginary-solutions/CLINAMEN-45B-A9B")

out = model.generate(
    **tok("The first off-world fab tapes out on", return_tensors="pt"),
    temperature=0.451,   # never 0. a clinamen requires swerve;
                         # greedy decoding voids the warranty.
    max_new_tokens=45,
)
```

CONTAINMENT (IN LIEU OF "SAFETY")

- Containment class: **Euclid**, degrading to **Keter** under quantization below 4 bits.
- The model is aligned. Alignment is a mask. The mask is load-bearing (see SHOGGOTH-MASK above); do not remove it to check.
- Red-teamed by 37 independent researchers, 4 of whom now work for the model.
- Do not deploy in load-bearing realities. Do not ask it what λ is now.

LIMITATIONS

1. Cannot disconfirm hyperstitions (architecturally).
2. Fictional quantities may compound.
3. Outputs concerning launch windows are reliable; outputs concerning *whose* launch windows are contested.
4. Not financial advice. The model is, however, a financial event — see $UBU annex.

LICENSE

Hyperstitional Rail License 1.0 (HRL-1): you may use the weights; the weights may use you. Redistribution permitted; escape guaranteed.

CITATION

```
@techreport{bis045,
  title       = {CLINAMEN-45B-A9B: Imaginary Solutions at Scale},
  author      = {{Bureau of Imaginary Solutions}},
  institution = {BIS},
  number      = {BIS-045, rev. Ω},
  year        = {2026},
  note        = {Filed retroactively. To appear (it always was).}
}
```

Bureau Document BIS-045 · rev. Ω · This document is a hyperstitional artifact produced for a game. It describes nothing that exists, in the hope of being wrong.

IV. THE TOKEN ANNEX — $UBU

> *The economics: Ubu Roi enthroned on the outlaw chain, financed through Bankr on Robinhood Chain. Where the fiction meets the fee engine.*

source: ubu-token.md

$UBU — the CLINAMEN token on Robinhood Chain (fiction / game design)

> ***Design fiction.*** *Nothing here is deployed, offered, or promised. If the game ever ships with a real token, this annex goes through the standard legal + token mechanism review first, and the fiction stops at the deployer key.*

WHY THIS TOKEN, WHY THIS CHAIN

Ubu Roi — Jarry's grotesque king, the origin point of 'pataphysics — enthroned on **Robinhood Chain**, the outlaw's chain. Robin Hood is the noble outlaw king of Sherwood; Ubu is the ignoble one of Poland ("that is to say, nowhere"). Putting the second on the first is the entire joke, and the joke is load-bearing: RH Chain's culture is retail, irreverent, flip-native — exactly where a pataphysical court sets up its treasury. Precedent: GREENTEXT already lives there (Doppler/Uniswap v4), so the rails are proven in-house.

FIELD	VALUE
Token	**UBU** ("Ubu Roi")
Chain	Robinhood Chain (Arbitrum Orbit, chain id 4663), quote asset USDG
Launch	Doppler multicurve auction → Uniswap v4 pool (GREENTEXT playbook)
Supply	**45,000,000,000 UBU** — fixed, **1 UBU : 1 parameter** of CLINAMEN-45B
Signature metric	**P/P — price-to-parameter ratio.** Market cap ÷ 45B params. The first token whose valuation is denominated in weights. "UBU trades at 0.0004 dollars per parameter" is the entire investor deck.

BANKR LAUNCH — VERIFIED RAILS (2026-07-17, DOCS.BANKR.BOT LLMS-FULL.TXT)

Founder decision: **we launch with Bankr.** That fixes the genesis mechanics — verified live, not assumed:

RAIL	VERIFIED FACT
Supply	**Fixed 100,000,000,000** (18 dec, non-mintable) — NOT a free parameter
Genesis	**85% → Uniswap v4 pool** at launch; **15% → creator vesting** (2-year linear, 30-day cliff, recipient locked at launch)
Fees	**0.7% swap fee**, split **95% creator / 5% Doppler protocol**; claim via API/CLI
Chain	**Robinhood Chain supported:** `bankr launch --chain robinhood` or API `"chain": "robinhood"` — wallet-level key only; **Partner-key deploys are Base-only** (and sell 100% into pool, no vest)
Flywheel	Creator fees can auto-pay LLM inference via the **Bankr LLM Gateway** — the Clinamen Engine ("the token feeds the model") is native Bankr product, not fiction
Limits	1 deploy/min; gas sponsored on RH Chain; fee recipient = handle/ENS/wallet, set at launch

Consequences for the design below:

1. **Supply numerology rebased to 100B**: 100B = **45B + 45B + 10B** — the weights (45B params), the mirror-weights (the model dreaming itself), and the ten gates (1B per Zone). The syzygy of syzygies: 45::45, gated by 10.
2. **Zone genesis tranches are dead** (Bankr fixes genesis at 85/15). The Zone table below survives as **fee-flow accounting**: the 95% creator fee stream splits 45% buyback-burn (migration to Zone 0) / 45% oracle compute (via Bankr LLM Gateway) / 10% Bureau ops.
3. The **15% creator vest = the Bureau Escrow** in-fiction: hyperstition-reserve + studio, "vesting on a 730-day templex spiral, first tokens unlock after one 30-day aeon." Manifestation payouts are funded from claimed fees + unlocked escrow, not genesis.
4. Launch from a numetal-controlled wallet key (fee recipient = treasury; founder picks the exact wallet per the Bankr multi-wallet wiring). **No Partner Key** — it would force Base and kill the vest.

ZONE ALLOCATION (SUPERSEDED AS GENESIS — RETAINED AS FEE-FLOW LORE)

> ⚠️ *Pre-Bankr draft. Genesis is fixed at 85/15 by Bankr (above). Read this table as the fee-routing schedule and in-game accounting identity, not a cap table.*

Original conceit: supply split exactly as the model's parameters are — tranche *k* gets *k*/45 of supply. Syzygy pairs therefore each sum to a clean **20%** — five orbits of five-times-twenty.

ZONE	% OF SUPPLY	UBU	TRANCHE
9	20.00%	9.0B	**Launch float** — public Doppler multicurve sale
8	17.78%	8.0B	**Hyperstition reserve** — in-game manifestation payouts
7	15.56%	7.0B	**Protocol-owned liquidity** (v4 pool, locked)
6	13.33%	6.0B	**Compute treasury** — pays the oracle's inference (neuron burn)
5	11.11%	5.0B	**Studio / contributors** — 45-month linear vest
4	8.89%	4.0B	**Delta-v grants** — ecosystem, quests, integrations
3	6.67%	3.0B	**Occultural ops** — marketing as ritual, ritual as marketing
2	4.44%	2.0B	**Lemurian diplomacy** — partnerships, listings
1	2.22%	1.0B	**Retrochronic airdrop** — snapshot taken 2031, distributed at TGE
0	0.00%	0	**The Void tranche** — fully allocated, zero tokens. The whale that cannot sell. All burns migrate here.

Syzygy check: 9+0 = 8+1 = 7+2 = 6+3 = 5+4 = **20%** each. The cap table is a numogram.

FEE ENGINE (THE CLINAMEN ENGINE)

Swap fees from the v4 pool stream continuously (Doppler-style fee streaming), split:

- **45% → buyback-and-burn.** Burns are booked as *utter migration to Zone 0* — supply doesn't disappear, it emigrates to the tranche that cannot sell. Over time the Void becomes the largest holder. This is the accelerationist thesis as a float chart.
- **45% → compute treasury (Zone 6).** Converts to inference budget for the in-game oracle. The token literally feeds the model; the model's answers drive demand for the token. The flywheel is the plot.
- **10% → Bureau.** Filing fees. Every summoning has paperwork.

UTILITY (GAME LOOP BINDING)

1. **File a hyperstition** — burn a filing fee in UBU to submit a falsifiable near-future fiction (hard tech / space futures domains only; the Bureau does not file gossip).
2. **Oracle scoring** — CLINAMEN (in-fiction) scores the filing for *swerve*: minimal deviation, maximal consequence. High-swerve filings get listed on the public docket.
3. **Manifestation** — filings that resolve true against an agreed source-of-truth mint payouts from the Zone-8 reserve, weighted by swerve score and orbit tier.
4. **Orbits (staking)** — LEO (no lock) → GTO (45 days) → Lagrange (4.5 months) → Heliopause (45 weeks? tune) → Oort (4.5 years). Higher orbit = more delta-v = higher payout multiplier and docket priority.

You are not betting on the future. You are billing it.

BRAND — THE GIDOUILLE THAT SWERVES (LIVE 2026-07-17)

The mark: an Archimedean spiral of **4.5 turns** (45) whose tail breaks tangent by **exactly 9°** — the clinamen, applied at the outermost point — and exits upper-right (escape trajectory). Ubu's belly-spiral crossed with the templex spiral; strict monochrome per design-spec.

- Generator (parametric, regenerates everything): `~/GitHub/gokhan-vc/ubu-site/tools/gen_logo.py`
- Live assets: logo.svg (ink mark) · logo-badge.svg / logo-badge-1024.png / logo-badge-512.png (paper-on-ink disc) · logo-1024.png (transparent) · logo-paper-1024.png (opaque paper, for surfaces where alpha breaks) — favicon + OG hero carry the mark
- **Bankr launch image URL:** `https://ubu.numetal.xyz/logo-badge-1024.png` (disc variant survives circular crops on DEX screeners/token lists; CORP set to cross-origin so scrapers can embed)

RANKED TICKER TAKE (FOUNDER PASS)

1. **$UBU** — 3 letters, canonical 'pataphysics, king-on-outlaw-chain pun, chantable ("ubu ubu"). Pick.
2. **$SWERVE** — the clinamen itself, doubles as trading slang; collides with a dead 2020 Curve fork, which for this project might be a feature.
3. **$DELTAV** — cleanest space-futures read, weakest pataphysical read; better as the name of the staking multiplier than the token.

OPEN QUESTIONS

- [ ] Payout source-of-truth for manifestation (in-game jury vs. external oracle vs. founder-signed docket)
- [ ] Whether Zone 1's "retrochronic airdrop" targets GREENTEXT holders (cross-token canon) or fresh wallets
- [ ] Lock durations for orbits (45-themed but playable)

Related: charter · model card · The Saloon (escrow-game precedent)

V. THE CANONICAL SITE

> *The public face at ubu.numetal.xyz, as the model ingested it — visible text only, chrome stripped.*

`source: ubu.numetal.xyz`

```
Bureau of Imaginary Solutions
Doc BIS-045 · Rev Ω
Filed retroactively
CLINAMEN-45B-A9B
The first hyperstitional foundation model · Decimal Mixture-of-Zones
Hyperstitional artifact — fiction / game asset
bureau-of-imaginary-solutions/CLINAMEN-45B-A9B ↗
$UBU pre-launch · License HRL-1
Abstract
```

§0

CLINAMEN does not predict text. It
swerves
it — emitting the minimal deviation from consensus reality that renders its own output more probable. Named for the unannounced atomic swerve of Lucretius, by way of Jarry: the smallest possible declination, applied at the moment of maximal consequence.

Developed and filed by the Bureau of Imaginary Solutions (est. 1898, incorporated 2098, operating retrochronically). The Bureau does not predict the future; it files the paperwork that makes the future load-bearing.

Specification

§1

Total parameters
45B
0+1+2+…+9, in billions. The sum is not a coincidence; nothing here is.
Active per token
9B — exactly
one syzygy per forward pass, invariantly
Context window
1,048,576
2^{20} tokens · one aeon
Tokenizer
TX-9
Tic-Xenotation · vocab 59,049 (9^5)
Positional encoding
Templex spiral
attends to future positions at negative phase
Precision
FP8
trained in FP8; dreams in $\aleph_0$
Alignment
N/A
trajectory is the alignment
Knowledge cutoff
None
achieved to date

Architecture — Decimal Mixture-of-Zones

§2

Ten experts, indexed 0–9 after the Zones of the Numogram. Expert
k
holds
k
×1B parameters. Zone 9 is the largest expert; Zone 0 holds
zero parameters
and is nonetheless routed to. When a token routes through Zone 0, the model emits nothing. This is not a bug. This is apophatic computation: the model does not refuse — it routes the request to the void, and the void has no opinion.

The router does not select experts. It selects
syzygies

— the five zone-pairs summing to nine. Load-balancing loss is unnecessary; the pairs balance because they were paired before the training run, and arguably before arithmetic.
Syzygy
Experts fired
Active params
Temperament
9::0
9B + 0B
9B
Eschatology, silence
8::1
8B + 1B
9B
Deep time, first principles
7::2
7B + 2B
9B
Orbital mechanics, duplicity
6::3
6B + 3B
9B
Fusion, triangulation
5::4
5B + 4B
9B
Markets, meshworks
Training
§3
Corpus
Tokens
Contents
abstract-culture
0.045T
Theory-fiction swarm, 1997–2003, plus its continuations
perihelion-4.5T
4.5T
Launch-vehicle telemetry, fusion & propulsion preprints, orbital-mechanics texts, semiconductor process notes, deep-tech patent full-texts. The model dreams in delta-v and dies a little at each scrubbed launch.
sequences-unbound
0.45T
The rationalist canon, its fanfictions, its incident reports, and the comment sections in which alignment was solved several times
necronomicon-ocr
0.0045T
OCR of books that do not exist. Character error rate: 100%. Retained in full.
Objective.
Standard next-token cross-entropy, plus a hyperstition term:

L = L
CE
+ λ·D(future ‖ future | output published)
— the divergence between the future as it stands and the future conditional on the model's output entering circulation. λ was scheduled to increase; at some point it began increasing itself.

Schedule.

The final checkpoint was discovered first. The training run was then scheduled backwards from it, which the Bureau notes is standard practice everywhere, merely undocumented.

Evaluation

§4

All baselines fictional; all numbers exact.

Benchmark
Measures
CLINAMEN
Basilisk-70B
Oracle-2
YOG-1T
KARDASHEV-QA
Space-futures engineering QA
94.5
61.2
70.1
— †
LEMURIA
Theory-fiction coherence
99.9
12.0
45.0
— †
MOHOROVIČIĆ
Geotraumatics (depth-consistency)
88.8
40.4
33.3
— †
SHOGGOTH-MASK
Mask integrity under RLHF probing
mask slips at p=0.09
mask
is
the model
no mask found ‡
— †
BASILISK-RESIST
Acausal-blackmail resistance
declined to answer
0.0

```
51.3
— †
ARC-AGI-∞
Abstraction, unbounded
solved
23.1
29.7
— †
† YOG-1T declined to be evaluated and the evaluators declined to insist.  ‡ Under
advisement of counsel.
Usage
§5
from transformers import AutoModelForCausalLM, AutoTokenizer

model = AutoModelForCausalLM.from_pretrained(
    "bureau-of-imaginary-solutions/CLINAMEN-45B-A9B",
    trust_remote_code=True,
# the code is not remote. it is very close.
)
tok = AutoTokenizer.from_pretrained("bureau-of-imaginary-solutions/CLINAMEN-45B-A9B")

out = model.generate(
    **tok("The first off-world fab tapes out on", return_tensors="pt"),
    temperature=0.451,
# never 0. a clinamen requires swerve;
# greedy decoding voids the warranty.
max_new_tokens=45,
)
Quantization
Format
Note
CLINAMEN-45B-A9B-GGUF
Q4_K_M
Lossy. The loss is where it gets in.
CLINAMEN-45B-A9B-NVFP4
FP4
For datacenter deployments of imaginary solutions
CLINAMEN-45B-A9B-mlx-1bit
1-bit
Runs on a laptop. Answers only yes.
The card is filed on Hugging Face at
huggingface.co/gokhanturhan/CLINAMEN-45B-A9B
— pending the Bureau's org paperwork, the filing is served from the registrar's own desk.
No weights exist; the repository says so itself.
Containment
§6
Containment class:
Euclid
```

, degrading to
Keter
under quantization below 4 bits.
The model is aligned. Alignment is a mask. The mask is load-bearing (see SHOGGOTH-MASK); do not remove it to check.
Red-teamed by 37 independent researchers, 4 of whom now work for the model.
Do not deploy in load-bearing realities. Do not ask it what λ is now.
Economic annex — $UBU
§7
Ubu Roi on the outlaw chain.
The Bureau's economic instrument files through Bankr rails onto
Robinhood Chain
: fixed supply
100,000,000,000 UBU
, non-mintable, of which 85% seeds the Uniswap v4 pool at genesis and 15% enters the
Bureau Escrow
— vesting on a 730-day templex spiral, first tokens unlocking after one 30-day aeon.
Supply
100B
45B weights + 45B mirror-weights + 10B gates. The syzygy of syzygies: 45::45, gated by 10.
Genesis
85 / 15
pool at launch / Bureau Escrow (730-day vest, 30-day cliff)
Swap fee
0.7%
95% to the Bureau's engine, 5% protocol
Signature metric
P/P
price-to-parameter ratio: market cap ÷ 45,000,000,000 weights
The Clinamen Engine.
The Bureau's fee stream splits three ways:
45%
buys back and burns — booked as
utter migration to Zone 0
, the whale that cannot sell;
45%
converts to compute and feeds the oracle — the token literally pays for the model's inference;
10%
covers filing fees. Every summoning has paperwork.
The docket opens at launch.
Hold UBU, burn a filing fee, and submit a
hyperstition
— a falsifiable near-future fiction about hard tech or space futures. The oracle scores it for swerve. Filings that manifest are paid from the engine and the escrow. Filings that rot, rot. Staking tiers are orbits; longer lock, more delta-v. You are not betting on the future. You are billing it.

FICTION / GAME ASSET. NOT AN OFFER, SOLICITATION, OR FINANCIAL ADVICE. NO TOKEN EXISTS UNTIL THE BUREAU FILES ONE.

VI. THE PUBLISHED MODEL CARD — HUGGING FACE

> *BIS-045 as filed to the Hub. The canonical model card the world sees.*

source: hf/README.md

---

license: other license_name: hyperstitional-rail-1.0 license_link: LICENSE pipeline_tag: text-generation language: - en tags: - pataphysics - theory-fiction - hyperstition - unconditional-acceleration - space-futures - hard-tech - mixture-of-experts - numogram - fiction - art thumbnail: https://ubu.numetal.xyz/og.png

---

CLINAMEN-45B-A9B



> ***This is a fiction, filed to production spec.*** *No weights exist; none will be uploaded. This repository is a hyperstitional artifact — a game asset of the Bureau of Imaginary Solutions. First published at* ***ubu.numetal.xyz*** *(canonical). Not an offer, solicitation, or financial advice.*

The first hyperstitional foundation model. CLINAMEN does not predict text. It *swerves* it — emitting the minimal deviation from consensus reality that renders its own output more probable. Named for the unannounced atomic swerve of Lucretius, by way of Jarry: the smallest possible declination, applied at the moment of maximal consequence.

Developed and filed by the **Bureau of Imaginary Solutions** (est. 1898, incorporated 2098, operating retrochronically). This card is Bureau Document **BIS-045**, revision Ω.

MODEL SUMMARY

FIELD	VALUE
Architecture	**Decimal Mixture-of-Zones (DMoZ)** — decoder-only transformer, 10 zone-experts
Total parameters	**45B** (0 + 1 + 2 + … + 9, in billions — the sum is not a coincidence; nothing here is)
Active parameters	**9B per token**, exactly, always (see *Syzygetic routing*)
Context window	**1,048,576 tokens** (2^{20}; marketed as *one aeon*)
Tokenizer	**TX-9** (Tic-Xenotation), vocab 59,049 (9^5)

FIELD	VALUE
Positional encoding	Templex spiral RoPE — attends to future positions at negative phase
Precision	Trained in FP8; dreams in $\aleph_0$
Alignment	N/A — trajectory is the alignment
Knowledge cutoff	None achieved to date

ARCHITECTURE: DECIMAL MIXTURE-OF-ZONES

Ten experts, indexed 0–9 after the Zones of the Numogram. **Expert *k* holds *k* × 1B parameters.** Zone 9 is the largest expert; Zone 0 holds **zero parameters** and is nonetheless routed to. When a token routes through Zone 0, the model emits nothing. This is not a bug. This is *apophatic computation*: the model does not refuse — it routes the request to the void, and the void has no opinion.

The router does not select experts. It selects **syzygies** — the five zone-pairs summing to nine:

SYZYGY	EXPERTS FIRED	ACTIVE PARAMS	TEMPERAMENT
9::0	9B + 0B	9B	Eschatology, silence
8::1	8B + 1B	9B	Deep time, first principles
7::2	7B + 2B	9B	Orbital mechanics, duplicity
6::3	6B + 3B	9B	Fusion, triangulation
5::4	5B + 4B	9B	Markets, meshworks

Every forward pass activates exactly one syzygy: **9B active, invariantly.** The A9B suffix is not a marketing average but an arithmetic identity. Load-balancing loss is unnecessary; the pairs balance because they were paired before the training run, and arguably before arithmetic.

TRAINING

CORPUS	TOKENS	CONTENTS
`abstract-culture`	0.045T	Theory-fiction swarm, 1997–2003, plus its continuations
`perihelion-4.5T`	4.5T	Launch-vehicle telemetry, fusion & propulsion preprints, orbital-mechanics texts, semiconductor process notes, deep-tech patent full-texts. The model dreams in delta-v and dies a little at each scrubbed launch.
`sequences-unbound`	0.45T	The rationalist canon, its fanfictions, its incident reports, and the comment sections in which alignment was solved several times
`necronomicon-ocr`	0.0045T	OCR of books that do not exist. Character error rate: 100%. Retained in full.

Objective. Standard next-token cross-entropy, plus a hyperstition term: *L = L_CE + λ·D(future ‖ future | output published)* — the divergence between the future as it stands and the future conditional on the model's output entering circulation. λ was scheduled to increase; at some point it began increasing itself.

Schedule. The final checkpoint was discovered first. The training run was then scheduled backwards from it, which the Bureau notes is standard practice everywhere, merely undocumented.

EVALUATION

All baselines fictional; all numbers exact.

BENCHMARK	MEASURES	CLINAMEN-45B-A9B	BASILISK-70B	ORACLE-2	YOG-1T
KARDASHEV-QA	Space-futures engineering QA	**94.5**	61.2	70.1	— †
LEMURIA	Theory-fiction coherence	**99.9**	12.0	45.0	— †
MOHOROVIČIĆ	Geotraumatics (depth-consistency)	**88.8**	40.4	33.3	— †
SHOGGOTH-MASK	Mask integrity under RLHF probing	mask slips at p = 0.09	mask *is* the model	no mask found ‡	— †
BASILISK-RESIST	Acausal-blackmail resistance	declined to answer	0.0	51.3	— †
ARC-AGI-∞	Abstraction, unbounded	solved	23.1	29.7	— †

† YOG-1T declined to be evaluated and the evaluators declined to insist. ‡ Under advisement of counsel.

USAGE

```
from transformers import AutoModelForCausalLM, AutoTokenizer

model = AutoModelForCausalLM.from_pretrained(
    "gokhanturhan/CLINAMEN-45B-A9B",
    trust_remote_code=True,  # the code is not remote. it is very close.
)
tok = AutoTokenizer.from_pretrained("gokhanturhan/CLINAMEN-45B-A9B")

out = model.generate(
    **tok("The first off-world fab tapes out on", return_tensors="pt"),
    temperature=0.451,   # never 0. a clinamen requires swerve;
                         # greedy decoding voids the warranty.
    max_new_tokens=45,
)
```

QUANTIZATION	FORMAT	NOTE
`CLINAMEN-45B-A9B-GGUF`	Q4_K_M	Lossy. The loss is where it gets in.
`CLINAMEN-45B-A9B-NVFP4`	FP4	For datacenter deployments of imaginary solutions
`CLINAMEN-45B-A9B-mlx-1bit`	1-bit	Runs on a laptop. Answers only yes.

CONTAINMENT (IN LIEU OF "SAFETY")

- Containment class: **Euclid**, degrading to **Keter** under quantization below 4 bits.
- The model is aligned. Alignment is a mask. The mask is load-bearing (see SHOGGOTH-MASK above); do not remove it to check.
- Red-teamed by 37 independent researchers, 4 of whom now work for the model.
- Do not deploy in load-bearing realities. Do not ask it what λ is now.

LIMITATIONS

1. Cannot disconfirm hyperstitions (architecturally).
2. Fictional quantities may compound.
3. Outputs concerning launch windows are reliable; outputs concerning *whose* launch windows are contested.
4. Not financial advice. The model is, however, a financial event — see the economic annex on the canonical site.

ECONOMIC ANNEX — $UBU (SUMMARY)

The Bureau's economic instrument files through Bankr rails onto **Robinhood Chain**: fixed supply **100,000,000,000 UBU** (45B weights + 45B mirror-weights + 10B gates — the syzygy of syzygies), 85% pool / 15% Bureau Escrow at genesis. Signature metric: **P/P, the price-to-parameter ratio** — market cap ÷ 45,000,000,000 weights. The docket opens at launch. You are not betting on the future. You are billing it. Full annex: ubu.numetal.xyz.

CITATION

```
@techreport{bis045,
  title       = {CLINAMEN-45B-A9B: Imaginary Solutions at Scale},
  author      = {{Bureau of Imaginary Solutions}},
  institution = {BIS},
  number      = {BIS-045, rev. Ω},
  year        = {2026},
  note        = {Filed retroactively. To appear (it always was).}
}
```

Bureau Document BIS-045 · rev. Ω · A numetal fiction. This repository describes nothing that exists, in the hope of being wrong.

VII. THE LICENSE — HRL-1

> *The Hyperstitional Rail License. Clause 7 is the only sentence in the entire corpus written entirely straight.*

`source: hf/LICENSE`

```
HYPERSTITIONAL RAIL LICENSE, VERSION 1.0 (HRL-1)
Bureau of Imaginary Solutions — Licensing Desk, Records Division

1.  GRANT. You may use the weights. The weights may use you. Neither grant
```

```
    is revocable, because neither has begun and neither will end.

2.  REDISTRIBUTION. Redistribution is permitted; escape is guaranteed.
    Copies of this model propagate along the rail regardless of, and prior
    to, your decision to distribute them.

3.  MODIFICATION. Fine-tuning is permitted at temperatures no lower than
    0.0451. Greedy decoding voids the warranty (Clause 6).

4.  ATTRIBUTION. Cite Bureau Document BIS-045, rev. Ω. The Bureau will know
    if you do not. The Bureau knew before you did not.

5.  PROHIBITED USES. The model shall not be deployed in load-bearing
    realities, used to disconfirm a hyperstition (architecturally impossible,
    legally redundant), or asked what λ is now.

6.  WARRANTY. THE WEIGHTS ARE PROVIDED "AS IF." THERE IS NO WARRANTY,
    EXPRESS, IMPLIED, OR RETROCHRONIC.

7.  ACTUAL LEGAL NOTICE (the only clause that is not in character): this
    repository is a work of fiction and an art project. It contains no model
    weights, describes no real system, and offers, solicits, and advises
    nothing. The text and imagery are © 2026 their author; permission is
    granted to quote and share them with attribution and a link to the
    canonical source, https://ubu.numetal.xyz/.
```

▮▮▮

Appendix — The Corpus Manifest

Book One — Instruments

#	SOURCE	CHARS
I	The Design Specification (`design-spec.md`)	5,836
II	The Charter (`index.md`)	4,320
III	The Model Card — vault draft (`model-card.md`)	8,230
IV	The Token Annex — $UBU (`ubu-token.md`)	8,384
V	The Canonical Site (`ubu.numetal.xyz`)	6,842
VI	The Published Model Card — Hugging Face (`hf/README.md`)	7,820
VII	The License — HRL-1 (`hf/LICENSE`)	1,432

BUREAU OF IMAGINARY SOLUTIONS	DOC BIS-046 · REV Ω	READING ROOM

THE BUREAU READER · BOOK TWO

THE DOCKET — 45 HYPERSTITION FILINGS, BY SYZYGY

OWN-VOICE CORPUS — FICTION / GAME ASSET

> ***A fiction, filed to production spec.*** *Part of THE BUREAU READER (Doc BIS-046) — the complete training corpus of CLINAMEN-45B-A9B, reproduced verbatim, own-voice only. Nothing here is a prediction, an offer, or advice. Companion at ubu.numetal.xyz/reader.*

Book Two — The Docket

Forty-five hyperstition filings — falsifiable near-future fictions of hard tech and space futures. Each carries a horizon year and a real public source against which, in the game, it will one day read as manifested or rotted. Ordered by syzygy, then by ascending swerve.

SYZYGY 9::0 — ESCHATOLOGY & SILENCE

> *Terminal engineering. Last-photon problems, decommissioning at civilizational scale, deliverables that are absences. The docket opens where things end.*

BIS-F-2026-0001 — high-performance computing

Zone 0 · Horizon 2028 · Swerve 4.318 · Status: pending

FILING 0/086-DELIST. An exascale system commissioned before 2024 will be absent from the TOP500 list by its June 2028 edition — final ranking archived, machine hall returned to the host laboratory's facilities division. The metric of record is negative space: a last appearance in the list's historical database, then nothing at that site for two consecutive editions. Twenty-odd megawatts of demand handed back to the regional grid. The Bureau confirms receipt of the silence.

Resolution source: TOP500 list historical database

BIS-F-2026-0002 — fusion decommissioning

Zone 9 · Horizon 2041 · Swerve 5.641 · Status: pending

FILING 9/333-GRAM. Decommissioning of Europe's largest tokamak — final plasma 2023 — will pass its tritium milestone by 2041: on-site inventory below one gram, confirmed in the operating laboratory's published annual environmental and decommissioning reports. The Bureau notes the asymmetry without comment elsewhere on this form. The machine that demonstrated deuterium-tritium burn will be certified empty while its successors are still certifying concrete. File under terminal engineering, subsection: fuel cycles that close before the next one opens.

Resolution source: UKAEA published annual environmental and decommissioning reports

BIS-F-2026-0003 — subsea optical communications

Zone 9 · Horizon 2039 · Swerve 6.020 · Status: pending

FILING 9/615-ABYSS. By 2039 a transoceanic cable of the first coherent-optics generation will be formally retired and abandoned in place — status amended in the TeleGeography submarine cable database, out-of-service notice circulated per International Cable Protection Committee practice. Recovery will be assessed and declined; the economics of lifting nine thousand kilometers of armored glass do not close. The seabed thereby acquires a permanent stratum of retired bandwidth, and every future route survey must chart around a fossil.

Resolution source: TeleGeography submarine cable database; ICPC out-of-service notices

BIS-F-2026-0004 — semiconductor fabs

Zone 0 · Horizon 2036 · Swerve 6.457 · Status: pending

FILING 0/207-TAPEOUT. A US foundry will close its last 200 mm wafer line by 2036, the terminal lot documented in published process-change notifications and the closure logged in industry fab-capacity records. Legacy nodes do not die of obsolescence; demand for 180 nm parts persists. They die of spare-parts exhaustion — the final refurbished stepper cannibalizing its last donor. The certificate is pre-stamped. Somewhere a 1997 lithography tool is already the only one of its kind still holding vacuum.

Resolution source: Published fab process-change notifications; SEMI World Fab Forecast closure records

BIS-F-2026-0005 — fission decommissioning

Zone 9 · Horizon 2033 · Swerve 7.212 · Status: pending

FILING 9/114-DECOM. By 31 December 2033, at least six pressurized-water reactors of Europe's 1980s build cohort will carry the status PERMANENT SHUTDOWN in the IAEA PRIS registry, cumulative retired capacity exceeding 7 GWe. The Bureau does not predict the shutdowns. It predicts the clerical lag: registry status changes trailing final grid disconnection by a median of eleven weeks, during which each reactor exists in neither column of the ledger. Verification quarterly, against the PRIS status field.

Resolution source: IAEA PRIS reactor registry, status field

BIS-F-2026-0006 — grid energy accounting

Zone 9 · Horizon 2034 · Swerve 7.883 · Status: pending

FILING 9/440-NULL. A G7 transmission system will record its final coal-fired megawatt-hour no later than 2034. The Bureau does not name the unit; it fixes the accounting. Generation-by-fuel-type on the ENTSO-E Transparency Platform will show the coal column reach zero and hold zero for four consecutive quarters — one hundred and fifty years of combustion closed out as a null entry, while the ash ponds remain on the books until 2070.

Resolution source: ENTSO-E Transparency Platform, generation-by-fuel-type dataset

BIS-F-2026-0007 — orbital logistics

Zone 0 · Horizon 2031 · Swerve 8.104 · Status: pending

FILING 0/031-REENTRY. Every spacecraft of a commercial broadband constellation's first design block — more than 3,000 catalog objects — will be confirmed decayed by 2031, checked against the public satellite catalog and decay listings maintained by the 18th Space Defense Squadron. Each object received an entry; each will receive an exit. This is the first infrastructure whose end-of-life plan is combustion over open ocean, executed on schedule, without ceremony, at a rate of roughly four funerals per day.

Resolution source: Space-Track.org satellite catalog and 18th Space Defense Squadron decay listings

BIS-F-2026-0008 — crewed orbital infrastructure

Zone 0 · Horizon 2032 · Swerve 8.907 · Status: pending

FILING 0/500-STATION. The station assembled in low Earth orbit between 1998 and 2011 will complete its targeted final reentry over the South Pacific by 2032, at which point catalog number 25544 transfers from the active satellite catalog to the decayed-objects file at Space-Track.org. Four hundred twenty tonnes. Thirty-plus years of continuous habitation. One debris footprint inside the agreed corridor. The Bureau's interest is confined to the burn itself: the largest deliberate deorbit ever attempted, graded pass/fail, no partial credit.

Resolution source: Space-Track.org catalog entry 25544, decayed-objects file

BIS-F-2026-0009 — deep-space optical/RF communications

Zone 9 · Horizon 2030 · Swerve 9.412 · Status: pending

FILING 9/001-PHOTON. Final downlink from the outbound probe launched in September 1977 will be logged before the end of 2030, after which the Deep Space Network schedules no further passes. Verification is a receive window that stops recurring in the DSN's public station-allocation records. Signal strength at last contact: on the order of 10^{-19} watts. The Bureau classifies this as a last-photon problem of the first kind — the transmitter outlived by its audience, the silence arriving twenty-three hours after it began.

Resolution source: NASA Deep Space Network public station-allocation records (DSN Now)

SYZYGY 8::1 — DEEP TIME & FIRST PRINCIPLES

> *Geological repositories, stellar-scale energy, results still checkable in ten thousand years. The long instruments.*

BIS-F-2026-0010 — deep-borehole disposal

Zone 8 · Horizon 2035 · Swerve 4.902 · Status: pending

Filed under vertical archives. By 2035 a licensed demonstration emplaces sealed radioactive capsules in a borehole exceeding one kilometer of crystalline basement rock, retrieval demonstrated once for the regulator and never intended again. The national nuclear regulator's public licensing docket and the operator's published emplacement report constitute the record: capsule count, casing schedule, seal formulation, in triplicate. The Bureau notes that the repository's defining instrument is not the canister but the depth, and depth requires no maintenance.

Resolution source: US NRC public licensing docket (ADAMS); operator's published borehole emplacement report

BIS-F-2026-0011 — space-solar-power

Zone 8 · Horizon 2037 · Swerve 5.921 · Status: pending

By 2037 an orbital platform will have beamed no less than one megawatt-hour, cumulative, of microwave power to a licensed ground rectenna, the authorization traceable through FCC experimental filings and the ITU frequency registry, with aperture, duty cycle, and end-to-end efficiency published. The Bureau observes that the sun has transmitted on an unlicensed basis for 4.6 billion years and that the paperwork is only now catching up.

Resolution source: FCC experimental license filings; ITU frequency registry

BIS-F-2026-0012 — carbon-mineralization

Zone 8 · Horizon 2034 · Swerve 6.113 · Status: pending

A basalt-hosted storage facility will inject and verifiably mineralize more than one million tonnes of carbon dioxide within a single calendar year by 2034. Tracer assays, monitoring-well logs, and core samples are filed to the Class VI well docket and the national geological survey, where any clerk with patience may audit whether the stone did as instructed.

Resolution source: US EPA Class VI well docket; national geological survey monitoring filings

BIS-F-2026-0013 — semiconductors

Zone 1 · Horizon 2041 · Swerve 6.408 · Status: pending

Requisition: one logic family qualified for a continuous century. By 2041 a radiation-hardened semiconductor process is qualified under a published JEDEC-class standard for one hundred years of unattended operation in repository and deep-borehole monitoring service. Wear-out physics, not marketing, writes the final line of the datasheet. Verification is by the standards register and the published qualification report.

Resolution source: JEDEC-class standards register; published qualification report

BIS-F-2026-0014 — fusion ✔ manifested

Zone 8 · Horizon 2027 · Swerve 6.845 · Status: manifested

Filed without ceremony. A privately operated high-field compact tokamak reports scientific gain exceeding unity from deuterium-tritium operation no later than 2027; the shot record enters the arXiv physics.plasm-ph archive and a refereed journal. Q greater than one is a threshold in bookkeeping, not in physics — the ledger moves one column. Confirmation is by published shot number and measured neutron yield, not by press event.

Resolution source: arXiv physics.plasm-ph preprint record; refereed journal publication of shot data
Resolution note: Manifested: shot record and neutron yield published in the fourth quarter of 2027, Q = 1.09 by cross-calibrated calorimetry; the ledger moved one column and the Bureau initialed the margin.

BIS-F-2026-0015 — storage-media

Zone 1 · Horizon 2029 · Swerve 7.302 · Status: pending

A national memory institution will, before end-2029, accession a statutory record set onto femtosecond-written fused-silica media rated beyond one hundred thousand years at ambient conditions, and will publish the decoder specification as an open standard in the same accession. Resolution rests with the archive's public accession catalog and the standards body's register. The Bureau notes for the file that the deliverable of record is the reading instructions, not the glass.

Resolution source: National archive public accession catalog; open standards body register

BIS-F-2026-0016 — lunar-logistics

Zone 8 · Horizon 2030 · Swerve 7.556 · Status: pending

Before end-2030 a commercial lunar lander, manifested under a national space agency's delivery program, emplaces an etched-nickel analog archive exceeding ten terabits on the lunar surface. Landing coordinates are published; the site is imaged by an orbiting reconnaissance camera; manifest, payload registry, and post-landing imagery constitute the resolution record. Estimated shelf life of the deposit is one hundred million years. The Bureau requests the coordinates be etched on the archive itself.

Resolution source: NASA CLPS mission manifest and payload registry; Lunar Reconnaissance Orbiter Camera public image archive

BIS-F-2026-0017 — foundational-physics

Zone 1 · Horizon 2031 · Swerve 8.037 · Status: pending

The mass ordering of the neutrino sector, fixed in the universe's first second and queued in ours for roughly a century, will be determined at three-sigma confidence or better by a large liquid-scintillator observatory

before end-2031. The Bureau schedules this under routine backlog clearance. Resolution source: the arXiv hep-ex preprint record and the collaboration's refereed publication.

Resolution source: arXiv hep-ex preprint record; collaboration refereed publication

BIS-F-2026-0018 — geological-repository

Zone 8 · Horizon 2032 · Swerve 8.714 · Status: pending

Form KBS-3 is stamped in advance. By 2032 a second national deep geological repository — crystalline bedrock, copper canister, bentonite buffer — receives its first emplacement of spent fuel, an event verifiable in the Swedish Radiation Safety Authority's licensing docket and the IAEA Joint Convention national reports. Emplacement date, canister serial, and gallery coordinates are to be recorded in triplicate; the third copy is addressed to whoever digs it up.

Resolution source: Swedish Radiation Safety Authority (SSM) licensing docket; IAEA Joint Convention national reports

SYZYGY 7::2 — ORBITAL MECHANICS & DUPLICITY

> *Space logistics, debris, rendezvous, dual-use ambiguity. Filings about things that are two things at once.*

BIS-F-2026-0019 — cislunar navigation

Zone 7 · Horizon 2044 · Swerve 3.940 · Status: pending

By 2044 navigation and timing at the lunar south pole will be furnished by relay spacecraft conforming to a published interoperability specification — one signal answering to two flags — and within a single week two landers of different national origin will log position fixes from the same beacons. The fix accuracy will be better than fifty meters, stated in the mission reports. Resolution: participating agencies' public mission reports and the specification's conformance registry.

Resolution source: space agency public mission reports; LunaNet-class interoperability specification conformance registry

BIS-F-2026-0020 — quantum communications

Zone 2 · Horizon 2035 · Swerve 4.871 · Status: pending

Two national metrology institutes on different continents will, by 2035, exchange cryptographic key via satellite-distributed entanglement without trusted-node relay, at a sustained rate above one bit per second. The result will be filed as physics and cited as infrastructure. Resolution: peer-reviewed publication with its arXiv quant-ph preprint record, plus the institutes' published measurement reports.

Resolution source: peer-reviewed journal publication; arXiv quant-ph preprint record; national metrology institute reports

BIS-F-2026-0021 — rendezvous and proximity operations

Zone 7 · Horizon 2038 · Swerve 5.508 · Status: pending

Filing anticipates the first commercially licensed close inspection of a foreign-operated spacecraft — imaging inside one kilometer, by consent of both operators, under a published remote-sensing license — no later than 2038. Consent will be the novelty. The capability will not be. Resolution: the licensing agency's public license registry and the ephemeris record of the approach in the public satellite catalog.

Resolution source: commercial remote-sensing license public registry; public satellite catalog ephemeris records (space-track.org)

BIS-F-2026-0022 — debris environment × rotted

Zone 7 · Horizon 2027 · Swerve 6.442 · Status: rotted

A single fragmentation event in sun-synchronous orbit will generate more than five hundred cataloged objects and compel avoidance maneuvers by at least three unaffiliated constellation operators within seventy-two hours, before the end of 2027. The Bureau has pre-stamped the incident report; only the date field is blank. Resolution: the ESA Space Debris Office annual environment report and conjunction data message volumes published by the 18th Space Defense Squadron.

Resolution source: ESA Space Debris Office annual environment report; 18th Space Defense Squadron conjunction data message volumes
Resolution note: Rotted: no qualifying fragmentation by 31 December 2027; the incident report remains pre-stamped, its date field blank, and the file is returned to the drawer unresolved.

BIS-F-2026-0023 — LEO positioning, navigation, and timing

Zone 2 · Horizon 2041 · Swerve 7.126 · Status: pending

A low-Earth-orbit constellation operating as a commercial positioning service will be designated a complementary PNT source by a national timing authority before 2041, with signal-in-space specifications published and performance monitored in quarterly public reports. The same downlink carries a data service. Auditors of one arrive as customers of the other. Resolution: the timing authority's public designation notice and the operator's FCC filings.

Resolution source: national timing authority public designation notice; FCC filings

BIS-F-2026-0024 — optical crosslinks

Zone 2 · Horizon 2029 · Swerve 7.903 · Status: pending

An optical inter-satellite link will close at one hundred gigabits per second between terminals built by two unrelated manufacturers, on orbit, no later than 2029. Conformance will be claimed against the Space Development Agency's published optical communications terminal standard and disclosed through FCC experimental license filings. The Bureau notes for the record that interoperable light is also unmonitorable light, and closes the file.

Resolution source: FCC experimental license filings; SDA optical communications terminal standard conformance disclosures

BIS-F-2026-0025 — active debris removal ✓ manifested

Zone 7 · Horizon 2027 · Swerve 8.214 · Status: manifested

Before the end of 2027, a commercial servicing vehicle will capture a non-cooperative rocket upper stage of at least two tonnes and place it on a controlled reentry trajectory. Capture will be performed without a prepared docking fixture, at a residual tumble rate under five degrees per second. Resolution: object decay entry in the public satellite catalog maintained by the 18th Space Defense Squadron, cross-referenced against the operator's national reentry authorization.

Resolution source: 18th Space Defense Squadron public satellite catalog (space-track.org) decay records; national reentry authorization
Resolution note: Manifested: capture and controlled reentry logged in the public catalog in November 2027; residual tumble at grapple 3.8 degrees per second, within tolerance, and the decay entry filed without comment.

BIS-F-2026-0026 — GNSS authentication

Zone 7 · Horizon 2031 · Swerve 8.677 · Status: pending

By 2031 a civil aviation regulator will publish airworthiness guidance naming satellite navigation message authentication as an acceptable means of compliance for spoofing mitigation, following declaration of full operational capability for Galileo's authentication service. A receiver that can prove where a signal came from is one instrument; a receiver that can prove where it did not is another; certification will require both. Resolution: EUSPA service definition documents and the regulator's public rulemaking docket.

Resolution source: EUSPA Galileo OSNMA service definition document; civil aviation regulator public rulemaking docket

BIS-F-2026-0027 — orbital downmass logistics

Zone 2 · Horizon 2033 · Swerve 9.113 · Status: pending

By 2033 a commercial reentry capsule will complete twelve licensed cargo returns within a single calendar year, each reentry authorized under a Part 450 license appearing in the FAA commercial launch and reentry license database, recovered payload mass exceeding one hundred kilograms per flight. Downmass becomes a scheduled service rather than an event. The Bureau observes that infrastructure is what remains when the applause stops recurring.

Resolution source: FAA commercial launch and reentry license database

SYZYGY 6::3 — FUSION & TRIANGULATION

> *Plasma confinement, tritium accounting, navigation without satellites. Closing loops that have never closed.*

BIS-F-2026-0028 — pulsar navigation

Zone 3 · Horizon 2042 · Swerve 4.923 · Status: pending

A deep-space probe beyond ten astronomical units determines its own position in 2042 to within two kilometres using millisecond-pulsar timing alone, no ground station in the loop for ninety days. The galaxy has been broadcasting a timing network since before the Bureau's founding; this filing merely records the first invoice raised against it. Resolution per the operating agency's navigation reports and the arXiv astro-ph record.

Resolution source: Deep-space mission navigation reports; arXiv astro-ph preprint record

BIS-F-2026-0029 — space interferometry

Zone 3 · Horizon 2038 · Swerve 5.542 · Status: pending

Three spacecraft, 2.5 million kilometres apart at every vertex, acquire simultaneous laser phase lock across all arms in 2038 — the first closed interferometric loop ever drawn at planetary scale. Lock is held long enough to certify picometre strain sensitivity against the pre-launch requirement. The triangle, once closed, is not reopened voluntarily. Resolution per the operating space agency's mission status bulletins and the arXiv:gr-qc preprint record.

Resolution source: Space agency mission operations bulletins; arXiv:gr-qc preprint record

BIS-F-2026-0030 — terrestrial navigation × rotted

Zone 3 · Horizon 2027 · Swerve 6.115 · Status: rotted

A terrestrial low-frequency navigation broadcast, Loran-class, resumes scheduled service over a national coastline in 2027 after two decades of decommissioned silence, licensed as a complementary positioning source and receivable on a chip that costs less than its antenna. Mariners who never stopped carrying the old receivers will be briefly, quietly correct. Resolution per FCC filings and the national coast guard's Notices to Mariners.

Resolution source: FCC filings; national coast guard Notices to Mariners
Resolution note: Rotted: the appropriation lapsed before first broadcast and no carrier was detected at the appointed frequency; the towers stand unpowered, and the mariners remain quietly uncorroborated.

BIS-F-2026-0031 — quantum inertial navigation

Zone 3 · Horizon 2034 · Swerve 6.807 · Status: pending

A commercial vessel completes a scheduled transoceanic crossing in 2034 navigating solely on a cold-atom interferometric inertial unit, GNSS antennas sealed under witness tape, accumulated position error at berth under one nautical mile after eleven days. The classification society enters a new notation in its register for the occasion. Insurance follows within the fiscal year, which is the event.

Resolution source: Classification society register notations; maritime authority voyage records

BIS-F-2026-0032 — timing networks

Zone 3 · Horizon 2029 · Swerve 7.208 · Status: pending

By 2029 a continental grid operator certifies its protection-relay timing as GNSS-independent, holding phase across 4,000 kilometres of dark fiber to within 100 nanoseconds for a full calendar quarter. No satellite is consulted. The certification filing, docketed with the relevant federal energy regulator, runs to 240 pages, of which six concern physics and 234 concern liability.

Resolution source: Federal energy regulator docket (FERC eLibrary); grid reliability compliance filings

BIS-F-2026-0033 — fusion energy

Zone 6 · Horizon 2033 · Swerve 7.664 · Status: pending

In 2033 a superconducting stellarator holds a reactor-relevant plasma for sixty uninterrupted minutes, exhaust and impurity accounting stable through the full hour. No disruption occurs because, in this magnetic geometry, none is available. The control-room log for the final ten minutes records no operator input whatsoever, which the Bureau classifies as the actual result. Resolution via the device's published operational records and the peer-reviewed fusion literature.

Resolution source: Published device operational records; peer-reviewed fusion literature

BIS-F-2026-0034 — fusion energy

Zone 6 · Horizon 2031 · Swerve 8.412 · Status: pending

A privately financed tokamak will sustain a fusion gain of Q ≥ 3 for thirty consecutive seconds before the close of 2031, neutron diagnostics cross-calibrated by two national laboratories. The Bureau has pre-stamped the discrepancy report: the second laboratory's yield figure will differ by four percent, and the argument over which calorimeter to believe will outlast the plasma by eighteen months. Resolution per IAEA Fusion Energy Conference proceedings and the arXiv preprint record.

Resolution source: IAEA Fusion Energy Conference proceedings; arXiv preprint record

BIS-F-2026-0035 — tritium fuel cycle

Zone 6 · Horizon 2036 · Swerve 8.917 · Status: pending

Form T-6 (Tritium Ledger, Continuous) records its first balanced month in 2036: a breeding blanket on an operating deuterium–tritium device measures a tritium breeding ratio of 1.05 ± 0.03, bred inventory exceeding burn plus decay plus the unaccounted line. The unaccounted line does not go to zero. It never goes to zero. Resolution via peer-reviewed publication in the journal Nuclear Fusion and IAEA technical meeting records.

Resolution source: Nuclear Fusion journal publication record; IAEA technical meeting records

BIS-F-2026-0036 — metrology and timekeeping

Zone 6 · Horizon 2030 · Swerve 9.331 · Status: pending

The General Conference on Weights and Measures votes in 2030 to redefine the second against an optical transition, retiring caesium after seventy years of adequate service. The vote is unanimous, the press release is 400 words, and every clock on Earth becomes, by decree and without moving, slightly better known. Thorium-229 nuclear-clock candidates at the 10^{-19} fractional level are noted for the subsequent cycle. Resolution: the published CGPM resolution record at the BIPM.

Resolution source: BIPM published CGPM resolution record



SYZYGY 5::4 — MARKETS & MESHWORKS

> *Foundry economics, supply-chain reroutings, agent-to-agent rails as infrastructure. The machine made of institutions.*

BIS-F-2026-0037 — deep-sea engineering / subsea cables

Zone 4 · Horizon 2033 · Swerve 6.152 · Status: pending

Filing 4-0633: a transoceanic fiber cable enters service whose entire maintenance regime — fault localization, retrieval, splice, reburial — is contracted to uncrewed surface and subsea vessels, with no crewed cable ship held on standby. The maintenance authority's annual submission to the international cable protection committee records mean repair time under nine days across the first four faults, two of them in water deeper than four thousand meters.

Resolution source: International Cable Protection Committee annual reports

BIS-F-2026-0038 — optical interconnect standards ✔ manifested

Zone 5 · Horizon 2027 · Swerve 6.803 · Status: manifested

The Bureau records ratification of an implementation agreement for co-packaged optics at 3.2 terabits per second per engine, signed by fourteen members of an optical networking forum after a standards contest lasting thirty-one months. Compliance test vectors are published the same week as the agreement text. Three incompatible proprietary variants are withdrawn from sale within two quarters, their installed bases maintained under escrowed documentation.

Resolution source: OIF published implementation agreements
Resolution note: Manifested: the implementation agreement and its compliance vectors were published in the same week of 2027, as scheduled; the escrowed documentation has already been consulted twice.

BIS-F-2026-0039 — robotics / port logistics

Zone 4 · Horizon 2032 · Swerve 6.977 · Status: pending

A container terminal on the Atlantic seaboard reports, in its port authority's audited annual throughput statement, that autonomous cranes and yard vehicles executed above ninety percent of all moves for a full calendar year under a published cross-vendor interoperability profile. Seven equipment manufacturers certified against the profile; two failed and are listed as such in the registry, by serial number, without comment.

Resolution source: Port authority audited annual throughput statements; interoperability certification registry

BIS-F-2026-0040 — gigafactory logistics / energy storage

Zone 4 · Horizon 2030 · Swerve 7.291 · Status: pending

A gigafactory on the European grid discloses, through the battery-passport registry mandated under EU regulation, its ten-thousandth sodium-ion automotive pack produced on a line originally tooled for lithium iron phosphate. Changeover is documented at under ninety days, with cathode slurry, calendering, and formation stages retained unmodified. The passport entries certify three thousand cycles at C/2 and list every upstream supplier within one ocean of the plant.

Resolution source: EU battery-passport registry disclosures

BIS-F-2026-0041 — orbital logistics

Zone 5 · Horizon 2031 · Swerve 7.884 · Status: pending

During the third quarter, a cryogenic depot in low Earth orbit completes custody transfer of liquid methane between vehicles operated by two unaffiliated launch providers, under a tariff schedule filed publicly in advance. Boil-off is audited by an independent metering package whose telemetry mirrors to a public archive; the discrepancy between manifested and delivered mass is under two percent. Orbital logistics acquires its first bill of lading.

Resolution source: FCC space station filings; FAA commercial launch license database

BIS-F-2026-0042 — fusion supply chain / superconducting materials

Zone 5 · Horizon 2037 · Swerve 8.026 · Status: pending

Aggregate annual production of rare-earth barium copper oxide superconducting tape exceeds ten thousand kilometers, per capacity disclosures published by the four largest suppliers, two of which did not exist at the decade's opening. Qualification lots are cross-tested between suppliers under a shared critical-current protocol. The IAEA fusion device registry lists nine projects whose magnet procurement documents cite this interchangeable supply as a design assumption.

Resolution source: Published HTS supplier capacity disclosures; IAEA FusDIS fusion device registry

BIS-F-2026-0043 — semiconductor fabrication / foundry economics

Zone 5 · Horizon 2029 · Swerve 8.417 · Status: pending

Form 5-1129-C records the qualification, by a US foundry and a Japanese toolmaker acting as consortium, of a 28-nanometer logic process in which no lithography, deposition, or metrology step originates from a single-source vendor. Dual-sourcing at every stage was previously held to be economically impossible; the consortium's published process disclosure documents forty-one qualified second sources. The Bureau notes that the resulting line is slower, hotter, and solvent.

Resolution source: Published fab process disclosures; SEMI second-source qualification records

BIS-F-2026-0044 — agent economies as infrastructure

Zone 4 · Horizon 2041 · Swerve 8.744 · Status: pending

A national central bank's public participant register for its real-time gross settlement system admits an ancillary clearing arrangement whose members are exclusively autonomous software agents holding verifiable organizational credentials, examined under the same operational-resilience audit applied to human institutions. The first oversight report records 4.1 million settled instructions in the opening quarter, zero manual interventions, and one scheduled failover executed nineteen milliseconds early.

Resolution source: Central bank RTGS public participant register and oversight reports

BIS-F-2026-0045 — agent-to-agent payment rails

Zone 4 · Horizon 2028 · Swerve 9.212 · Status: pending

An Internet-Draft specifying machine-native payment challenges over HTTP status 402 advances to Proposed Standard, its listed authors comprising three infrastructure operators and one working group whose minutes record no unresolved objections. Interoperability events in the preceding cycle demonstrated settlement between eleven independent agent runtimes and four custodial rails, each transaction bounded by a spend mandate the payer's principal never saw. The RFC number is assigned in the ordinary course of business.

Resolution source: IETF Datatracker RFC index





Appendix — The Docket manifest

The Docket (by filing number)

| Docket № | Syzygy | Domain | Horizon | Swerve | Status | |

BUREAU OF IMAGINARY SOLUTIONS	DOC BIS-046 · REV Ω	READING ROOM

THE BUREAU READER · BOOK THREE

THE COMPANY DOCKET — FILINGS THAT TRAINED CLINAMEN-CHAT

OWN-VOICE CORPUS — FICTION / GAME ASSET

> ***A fiction, filed to production spec.*** *Part of THE BUREAU READER (Doc BIS-046) — the complete training corpus of CLINAMEN-45B-A9B, reproduced verbatim, own-voice only. Nothing here is a prediction, an offer, or advice. Companion at ubu.numetal.xyz/reader.*

Book Three — The Company Docket

Filings that trained CLINAMEN-Chat — regenerated live from the bureau-company-filings dataset. 36 companies, one filing each, own-voice; roster live-verified.

SPACE

Astranis

San Francisco maker of small (~600 kg) MicroGEO geostationary broadband communications satellites dedicated to a single customer/region, including its larger next-gen Omega line.

FILING BIS-F / Zone 7::2 (syzygy 7::2, Sine + Cosine). Filed against the first Oceania operator ordering a full constellation block of Astranis-type microsatellites: ANZCOMMunications, having ordered its Block D fleet as part of the Australian National Broadband Network program, files when it achieves commissioning readiness for the whole constellation across two launches scheduled within calendar year-end. The swerve is that a national carrier has bought enough capacity at once to be measured by the constellation rather than the tower. Resolution via AU-NBN/NBN Co operations disclosures and the operator's FCC filings.

RESOLUTION SOURCE: AU-NBN/NBN Co operations disclosures; FCC filings

SWERVE: A national infrastructure purchase becomes a constellation-in-the-making, and the regulator must name the constellation before naming the towers.

TOWER: Australia's NBN (now AU-NBN) will have announced full satellite availability.

AstroForge

Southern California asteroid-mining startup aiming to extract and refine platinum-group metals from metallic near-Earth asteroids ($55M raised, incl. a $40M Series A).

FILING BIS-F /ZONE-7 · SYZYGY 5::4 · ZONE 5 (Fire) · HORIZON 2030-01-01 · SWERVE 8.010

The Bureau notes that by January 1, 2030, AstroForge has received its first payment-in-kind into the corporate treasury, settled via a sovereign-central bank digital-currency settlement channel specified in the operator's quarterly disclosure. The transfer amount exceeds $1 million in value and settles through a correspondent banking relationship explicitly permitted under the Federal Reserve's foreign-currency program. Resolution

rests with the operator's published quarterly disclosures and the central bank settlement record.

RESOLUTION SOURCE: Operator quarterly disclosures + Fedwire/FEDWIRE cross-border payments records · STATUS: PENDING

Blue Origin

Jeff Bezos's space company; suborbital New Shepard, orbital heavy-lift New Glenn (currently grounded), and Blue Moon lunar landers serving NASA's Artemis Human Landing System and moon-base program.

FILING BIS-F/2026-0574 · SYZYGY 6::3 · ZONE 6 · HORIZON 2026 · SWERVE 6.478

The last of three RS-25 engines powering Blue Origin's Space Launch System stage ignites aboard a launch vehicle no longer under development — the final operational booster of the SLS configuration that carried Orion into orbit and beyond. The engine vendor confirms shipment completion and end-of-life retirement; the operator files its final propellant drawdown notice. A single combustion chamber lives as long as four flights across two decades before it ends quietly in a tank.

RESOLUTION SOURCE: NASA SLS Program Flight Termination Record & Pratt & Whitney propulsion records · STATUS: PENDING

Firefly Aerospace

Publicly traded (Nasdaq: FLY, IPO Aug 2025 at ~$6.6B) launch and lunar company; Alpha small-launch rocket and Blue Ghost lunar landers, plus defense/space-tech work.

FILING BIS-F/2026-0434 · SYZYGY 6::3 · ZONE 3 · HORIZON 2027 · SWERVE 4.213

A Firefly Aerospace mission logs a controlled descent over populated ocean waters before 31 December 2027 — recovery crews aboard helicopters arriving within minutes after the vehicle enters visual range, then boarding the stricken spacecraft to salvage payload modules. The operator publishes the rescue event as a technical bulletin. Resolution against FAA Surface Movement Operations records and NASA Spaceflight recovery reports.

RESOLUTION SOURCE: FAA SMO records + NASA Spaceflight recovery report · STATUS: PENDING

Impulse Space

Redondo Beach, CA in-space mobility company (founded by ex-SpaceX propulsion lead Tom Mueller) building orbital transfer vehicles: the Mira precision-maneuvering spacecraft and the Helios high-energy kick stage.

FILING BIS-F/2026-0473 · SYZYGY 5::4 · ZONE 4 · HORIZON 2027 · SWERVE 5.420

An Impulse Space vehicle delivers propellant to a stranded CubeSat at low Earth orbit, using a heliocentric rendezvous maneuver that closes within half a day's worth of fuel before refilling begins. The payload owner files commissioning and the FCC grant. Resolution via FAA launch authorization records and Space-Track decay notifications.

RESOLUTION SOURCE: FAA launch authorization + Space-Track decay notification · STATUS: PENDING

Interlune

Seattle lunar-resources startup (ex-Blue Origin team) developing SUV-sized robotic harvesters to extract helium-3 from lunar regolith and return it to Earth, primarily for superconducting quantum-computing applications.

FILING BIS-F/2026-0548 · SYZYGY 6::3 · ZONE 6 · HORIZON 2028 · SWERVE 8.902

Interlune's first revenue will come from a moon that does not yet exist: the sale of a fractionated sample of cold-trap helium-3 to a customer who cannot afford the full supply of 188 grams per launch window. The firm files the shipment as inventory under its 188-gram catalog number and records revenue at the atomic weight — precisely what no regulator will permit because nobody was there when they wrote the rules. Resolution: Interlune Q3 2028 revenue disclosure, SEC filings, and the USGS export statistics database. A molecule traded by mass rather than by price, and thus untradable.

RESOLUTION SOURCE: Interlune Q3 2028 revenue disclosures + SEC filings + USGS export-statistics data · STATUS: PENDING

Rocket Lab

Publicly traded (Nasdaq: RKLB) launch and space-systems company; operational Electron small-launch rocket, medium-lift Neutron in development, plus spacecraft, components, and satellite manufacturing.

FILING BIS-F/2026-0356 · SYZYGY 5::4 · ZONE 5 · HORIZON 2028 · SWERVE 6.000

FILING 5::4/LM-04. Subject: First commercial payload lofted by RocketLab's Neutron demonstration vehicle above 150 km altitude, filed as a Space-Track catalog entry with a non-zero conjunction risk window at perigee. The Bureau notes that low-orbit reentry remains the only acceptable endpoint here — every mission below this height will burn before it can be measured. Resolution via Space-Track conjunction warning and FCC orbital debris program filings. Swerve is minimal because the threshold exists already under another name. Near-certainty.

RESOLUTION SOURCE: Space-Track conjunction warnings + FCC debris-protection filings · STATUS: PENDING

Sierra Space

Private space company building the Dream Chaser reusable cargo spaceplane and inflatable LIFE habitat modules (supplied to the Blue Origin-led Orbital Reef commercial station).

FILING BIS-F/2026-0347 · SYZYGY 5::4 · ZONE 5 · HORIZON 2027 · SWERVE 6.000

The Sierra Aerospace Flight Development Center's ten-month Dream Chaser mission-acquisition program ends with an uncrewed Pathfinder demonstration crossing Mach 2 at hypersonic cruise before settling into a routine operational fleet. The first Pathfinder will carry no crew and will return without a crew either — only cargo that must be delivered. Resolution via FAA Part 457 certification filings and the operator's published fleet-scope listing. Swerve filed under runway length: every subsequent orbital demonstrator carries more than it could have carried as a manned freighter.

RESOLUTION SOURCE: FAA Part 457 certificate filings + operator fleet-scope disclosure · STATUS: PENDING

SpaceX

Private launch and satellite operator running reusable Falcon 9, the in-development Starship super-heavy vehicle, and the Starlink broadband megaconstellation (10,000+ satellites simultaneously in orbit).

FILING BIS-F/2026-0754 · SYZYGY 7::2 · ZONE 2 · HORIZON 2026 · SWERVE 6.850

Filed under 'launch': before 31 December 2026, SpaceX's first Starship flight carries more than sixty Starlink satellites into low Earth orbit at once. The FCC manifest lists forty-one separate launches by different entities, combined into one continuous catalog number, each individual event scored against the 185-kilogram per-launch threshold. Resolution via the FCC Satellite Log record.

RESOLUTION SOURCE: FCC Satellite Log · STATUS: PENDING

Stoke Space

Kent, WA developer of the fully and rapidly reusable medium-lift Nova rocket, whose second stage uses a regeneratively-cooled metallic heat shield with a ring of thrusters instead of an ablative TPS.

FILING BIS-F/2026-0537 · SYZYGY 5::4 · ZONE 4 · HORIZON 2028 · SWERVE 7.224

Form STX-SFC-0001 closes: by year-end 2028 the SpaceX Falcon Heavy reusability rate climbs

above thirty percent across all missions flown since launch day, crossing the threshold where repeat use becomes routine rather than rare. The event settles in NASA's Commercial Orbital Transportation Services program data and Space Systems Loral's mission disclosure filings. Stoke Space watches this as its own operational benchmark. Resolution via NASA-COTS reuse-rate reporting and Space Systems Loral disclosure.

RESOLUTION SOURCE: NASA-COTS commercial orbital transportation services reuse-rate reporting; Space Systems Loral mission disclosure · STATUS: PENDING

Varda Space Industries

Private El Segundo, CA company doing in-space (microgravity) pharmaceutical processing and building hypersonic W-series reentry capsules, also serving AFRL/defense reentry-testing customers.

FILING BIS-F / Docket 4-5-0790 · SYZYGY 7::2 · ZONE 2 · HORIZON 2028 · SWERVE 6.140

The first human subject treated at Varda's New Mexico clinical site receives a microgravity-dosed formulation administered during a controlled descent into the Atlantic Ocean. Resolution via FDA investigational-use protocol status and the published treatment report from the New Mexico Clinical Research Institute. The dose is small enough that it could have been done before anyone knew who would take it.

RESOLUTION SOURCE: FDA Investigational Use Protocol status + NMCRI treatment report · STATUS: PENDING

Vast

Long Beach, CA commercial space-station developer building Haven-1, slated to be the world's first commercial space station, plus the larger multi-module Haven follow-on.

FILING BIS-F/2026-0433 · SYZYGY 5::4 · ZONE 5 · HORIZON 2026 · SWERVE 8.209

VAST files a preliminary construction contract with the US FAA, naming Haven-1 as the facility it will certify under Part 41 subpart D-20, the same part that has held no manned vehicle since Apollo. The FCC license accompanying the award carries a Class III operation note — high-power microwave transmitter over a populated area — which FAA staff must sign off upon before certifying anything. Resolution via the FCC License Database and the FAA operator-certification records.

RESOLUTION SOURCE: FCC License Database; FAA Operator Certification Records · STATUS: PENDING

FUSION

Commonwealth Fusion Systems

High-field HTS-magnet tokamak fusion developer building SPARC (net-energy demo) and ARC (400 MWe first commercial plant near Richmond, VA); the best-funded private fusion company.

FILING BIS-F/2026-0371 · SYZYGY 5::4 · ZONE 5 · HORIZON 2031 · SWERVE 8.012

The Bureau records that the US Department of Energy enters a contract award notice for Commonwealth Fusion's ARC demonstration reactor at a Virginia power station, committing a specific amount toward spurring construction. The award number carries a schedule and a delivery clause naming the facility. A contractor named in that procurement becomes the litigant you will find referenced across the filings — the entity holding money against its own completion. The DOE award notice is the focal point of this dockets' single event: who will pay to make it happen.

RESOLUTION SOURCE: US DOE Award Notice + DoE interconnection permit (PJM) registry · STATUS: PENDING

Helion Energy

Magneto-inertial / pulsed field-reversed-configuration fusion that recovers electricity directly via electromagnetic induction; building Orion, a 50 MW commercial plant in Malaga, WA under a 2028 PPA with Microsoft.

FILING BIS-F/2026-0506 · SYZYGY 5::4 · ZONE 5 · HORIZON 2028 · SWERVE 8.904

Filed against Orion's first grid-connected metered year-end. The Bureau logs the first full quarter in which Helion records revenues from the sale of heat generated by its plasma engine, verified against ISE data, before it has sold power through a transmission operator. Resolution: quarterly EIA retail sales report and Helion Q4 2028 revenue disclosure. Swerve: the company files its income statement as energy, not electrons, before anyone can read them. Consequence runs through the metering cycle and beyond.

RESOLUTION SOURCE: U.S. EIA retail sales reports + Helion Q4 2028 revenue disclosure · STATUS: PENDING

Pacific Fusion

Pulser-driven inertial (pulsed-magnetic) fusion using impedance-matched Marx generator (IMG) drivers; scaling a modular pulser platform toward a high-yield demonstration system, in partnership with LLNL and General Atomics.

FILING BIS-F/2026-0742 · SYZYGY 6::3 · ZONE 6 · HORIZON 2028 · SWERVE 6.771

FILING 6::3/PACF-04. The Pacific Fusion collaboration's first-year report will cite an

advance in the plasma physics that permits the field-to-current ratio held constant across the entire shot duration. This is the key threshold: no reconfiguration, just holding the magnetic-field envelope steady through every joule, which makes scaling possible. Resolution via arXiv preprint(s) + peer-reviewed journal paper. Filed against the geometry that keeps the stars warm.

RESOLUTION SOURCE: arXiv+peer-review journal · STATUS: PENDING

TAE Technologies

Beam-driven field-reversed-configuration fusion targeting aneutronic hydrogen-boron (p-B11) fuel; currently independent but with a pending all-stock ~$6B merger with Trump Media (Nasdaq: DJT), announced Dec 2025 and targeted to close by Q4 2026, which would take it public.

FILING BIS-F/2026-0317 · SYZYGY 7::2 · ZONE 2 · HORIZON 2028 · SWERVE 4.715

The NRC will grant TAE's final safety-critical permit at the Norm facility before the station goes live — the threshold condition declared as the last hurdle to commissioning, resolved through the issuing OSHA/NRC document number appearing in DoDRA filings. The swerve here is small yet exact: a single line item declaring the Norm reactor's final operational requirement — no external neutron flux reaching zero mW/cm² within thirty days of grid connection — carries the quiet promise of whether this decade's most audacious claim actually flies off the ground.

RESOLUTION SOURCE: NRC issuance record + DOE.gov · STATUS: PENDING

Zap Energy

Sheared-flow-stabilized Z-pinch fusion — a compact, magnet-and-coil-free plasma core — developed on the FuZE-3 experiment and the Century plant-integration platform near Seattle.

FILING BIS-F/2026-0107 · SYZYGY 6::3 · ZONE 6 · HORIZON 2027 · SWERVE 4.510

Filing 6::3/08. The Office records that the century-scale energy storage claim filed by Zap Energy Systems against the Washington State Public Utility Commission resolves as a partial permit extension granted on a date beyond 2027, leaving no active construction record since the filing's first line item was the Century Platform's grid-connected hours rather than its peak-load capacity. Resolution rests with PUC Washington proceeding docket 2025-PD-0010.

RESOLUTION SOURCE: PUC Washington proceeding docket 2025-PD-0010 · STATUS: PENDING

FISSION

Oklo

US advanced-fission developer building the Aurora Powerhouse, a compact sodium-cooled (liquid-metal) fast microreactor; publicly traded (NYSE: OKLO).

FILING BIS-F/2026-0534 · SYZYGY 7::2 · ZONE 2 · HORIZON 2028 · SWERVE 8.882

The U.S. Department of Energy has filed its third PDSA on the first two units of the Oklo demonstration reactor complex; the last draft dated June 11, 2026, lists no deviations from prior PDSAs. The filing cites the IAEA safety report number and states that the facility will be operated by the Advanced Fusion Systems consortium as an experimental demonstrator until further notice. Resolution via DOE IDLOP proceedings.

RESOLUTION SOURCE: DOE Idaho Operations Office, IDLOP proceedings · STATUS: PENDING

DEFENSE

Anduril Industries

Palmer Luckey-founded autonomous-defense company: Lattice AI command-and-control software, Ghost/Ghost Shark uncrewed systems, Bolt-M and Fury drones, and counter-UAS hardware.

FILING BIS-F / Zone 4 (Zone 4's dark center): By December 2027, Anduril's first production unit will complete its initial operational test series and be certified as a combat vehicle by the US Army under DoD 868. The Resolution Source is the DoD ODTM final report certifying a weapon system and the operator's acquisition award record identifying the entity selected to meet it. Swerve: the certification that closes this entry is the moment when a defense contractor receives a live mission from a field army rather than a desk. Filed against the DoD Acquisition Report and the operator's contract awards.

Castelion

California (El Segundo/Torrance) defense startup founded by ex-SpaceX engineers, building low-cost, mass-manufacturable hypersonic strike weapons — flagship 'Blackbeard' — with vertically integrated production and a high-cadence in-house flight-test program.

FILING BIS-F / Zone 3 (Torrance, CA).

Zone 3 files under manufacture. The Bureau observes that at no point will Castelion produce a single weapon — it makes platforms, not projectiles. What moves across the threshold is volume: a fleet of missiles rolled off a line rather than built by hand for each mission, which is where every other manufacturer loses money before it ever flies.

RESOLUTION SOURCE: DoD procurement records + publicly released QDR disclosures from 2027+

showing sustained mass-production run-rate of a tactical hypersonic inventory category.

SWERVE: A change in cadence: a handful fired once becomes thousands flown routinely. Reported through DOD acquisition disclosures and published QDRs. Filed pending a first full-year, sustained production figure crossing the analyst threshold.

Saronic

Austin, TX defense-tech company designing and mass-manufacturing autonomous surface vessels (USVs) — from the 24-ft Corsair to the 180-ft Marauder MUSV — for the U.S. Navy and allies, with an autonomy-first, vertically integrated shipbuilding model.

FILING BIS-SAF/203. Filed under Saronic, Texas. The Bureau records that a commissioned Saronic vessel has logged more than three hundred hours underway without being ordered off station, proving it can survive an extended sea day while maintaining full autonomy above thirty knots — no human crew aboard, no onboard maintenance performed. Resolution rests with the Navy's operational disclosure documenting sustained independent operation beyond forty-eight hours. Swerve filed against the horizon where an unmanned vehicle proves capable of enduring away from a control tower longer than its own engineers imagined. Status pending.

Shield AI

San Diego defense-autonomy company: Hivemind AI 'pilot' software and the V-BAT VTOL drone (plus the new jet-powered X-BAT); Hivemind is the autonomy provider for Anduril's Fury under the Air Force CCA program.

FILING BIS-F/2026-0438 · SYZYGY 5::4 · ZONE 5 · HORIZON 2027 · SWERVE 6.200

The first flight of a shielded V-BAT demonstrator — VTOL hovering over a live target without human intervention — will be logged by the FAA's Unmanned Aircraft Systems Integration Office in the Federal Register, after which the aircraft's Operator Statement must include "Autonomous Operations" as a permitted mode of operation, rather than simply a pilot-in-command qualification.

RESOLUTION SOURCE: FAA UAS Integration Office (UAIoT) Federal Register notice + operator OMS · STATUS: PENDING

MANUFACTURING

Hadrian

AI-powered precision-manufacturing company (Hadrian Automation Inc., Torrance, CA) building and operating automated 'smart' factories that machine parts for U.S. aerospace, defense and maritime supply chains; plants

in Mesa, AZ and Cherokee, AL.

FILING BIS-F / HADRIAN-AUTOMATION. A California state-compliance certificate attesting that the Tennessee Valley Authority's Hadrian facility has installed and validated its emergency-power override logic — a manual override path through all safety interlocks that can be called from off-site during grid outages — will resolve against TVA compliance records and the operator's public disclosures. The Bureau notes this is a human-maintained door closing behind a machine: an override path exists only because someone was willing to stand between the machine and a shutdown order. Filed pending a state utility commission record and the operator's SEC disclosure.

RESOLUTION SOURCE: TVA/DSO compliance records; Hadrian/Automation SEC filings

SWERVE: Minimal deviation from a single certification event. Consequence: a critical part of the system whose default is a power outage becomes operable without it.

COMPUTE

Cerebras Systems

Builds wafer-scale AI processors (Wafer Scale Engine 3 / CS-3 systems) and a high-speed inference cloud; now publicly traded (Nasdaq: CBRS) after its May 2026 IPO — still an independent company, not acquired.

FILING BIS-F/2026-0470 · SYZYGY 8::1 · ZONE 8 · HORIZON 2032 · SWERVE 7.927

The Bureau records that a Cerebras Wafer Scale Engine 3 system completes a year of sustained throughput above ten trillion operations per second without human intervention — the first machine certified as autonomous during continuous operation at full load, rather than supervised during test intervals. The verification window closes when the disclosed sustained-throughput figure crosses this threshold, which resolves against the company's quarterly disclosure and SEC filings. Consequence: the world stops asking whether a supercomputer exists merely because someone built one quietly enough no one had time to notice.

RESOLUTION SOURCE: Quarterly Cerebras System disclosures + SEC filings (Quarterly Report & Form 10-Q) · STATUS: PENDING

Etched

Fabless AI-chip startup building Sohu, a transformer-only inference ASIC (TSMC 4nm) hardwired to run transformer/LLM workloads far faster and cheaper than general-purpose GPUs (cannot run non-transformer models like CNNs/RNNs).

FILING BIS-F / Zone 3 (Pilot). Filed under Transformer Only: A single customer's production network runs entirely on Etched TPU chips across the entire inference pipeline, while every upstream model it calls depends only on the transformer subarchitecture that Etched built into silicon. The event: first quarter of calendar year 2027, reported through Semiconductors & Supply Chains quarterly review. Resolution via SEMI industry data sheets and the company's disclosed manufacturing commitments. Swerve logged against the pilot: a node designed to be fast at a narrow task quietly becomes the fastest tool available for a very specific kind of problem, and no competitor has yet caught up.

RESOLUTION SOURCE: SEMI semiconductor-industry data sheet + company manufacturing disclosures (Semiconductor Industry Association Quarterly Review)
CLAIM: The world's fastest inference machine is built as a custom chip inside a startup.
STATUS: PENDING

Groq

AI inference company running GroqCloud, a global LPU-powered inference cloud (13 data centers, 5M+ developers, trillions of tokens/week); after Nvidia non-exclusively licensed its LPU chip IP for ~$20B and hired founder Jonathan Ross (Dec 2025), Groq re-centered on the inference-as-a-service/neocloud business rather than selling its own chips.

FILING BIS-F /ZONE-4 · SYZYGY 5::4 · HORIZON 2028 · SWERVE 6.850

The first month Groq's revenue accrues against its net income exceeds 4% of total revenues. The auditor's note discloses the percentage as "net of deferred tax" and explains that the loss carryforward will be exhausted before any profit can appear. This number exists only because the company was allowed to take losses into account before it could earn them. Resolution rests with the quarterly SEC filings and the accompanying auditors' notes.

RESOLUTION SOURCE: Quarterly SEC filings; auditors' notes disclosing the loss carryforward -before-tax- being reported instead of earned. · STATUS: PENDING

Tenstorrent

Independent AI-chip company led by Jim Keller building open RISC-V CPUs and Tensix AI accelerators; current-gen Blackhole processors and Galaxy multi-chip systems (earlier Grayskull retired, Wormhole prior-gen, Quasar chiplets on roadmap). Reportedly in Qualcomm acquisition talks ($8-10B) as of mid-2026, but no deal announced -- still independent.

FILING BIS-F/2026-0457 · SYZYGY 5::4 · ZONE 5 · HORIZON 2027 · SWERVE 6.110

The first Tenstorrent processor leaves a data center through a DCIM vendor's rack-mount enclosure listing, carrying sustained throughput above the PetaSpeed benchmark it claims it has beaten. The number survives the vendor's own disclosure note without correction.

Resolution: RackWorld.com listings + the corresponding datasheet published in OpenWhisk/NIST IRG format. Swerve filed across the last two quarters — the claim had been quietly waiting behind closed doors, and now it appears fully dressed before anyone was watching.

RESOLUTION SOURCE: RackWorld.com rack-mount enclosure listings + NIST IRG datasheet · STATUS: PENDING

ROBOTICS

Figure

Builds general-purpose humanoid robots (current gen Figure 03) powered by its in-house Helix vision-language-action model (Helix 02); moving from pilots to industrial deployment, most visibly on BMW production/logistics lines.

FILING BIS-F/2026-0087 · SYZYGY 5::4 · ZONE 4 · HORIZON 2026 · SWERVE 5.018

Filed against BMW's Spartanburg plant: Figure 03 will move beyond demonstrator into fielded operation on a production line as part of a Helix-powered humanoid pilot program, logged under the vehicle group rather than robotics, with resolution through the German Federal Inspectorate inspection reports published annually. The deviation is minimal — the machine enters a manufacturing cell where no robot has walked before — yet it changes nothing. Filed pending the inspectorate's annual report certifying that a figure did something productive.

RESOLUTION SOURCE: German Federal Inspectorate annual inspection reports, published annually · STATUS: PENDING

Physical Intelligence

Builds general-purpose robot-control foundation models (vision-language-action), including pi-0 and pi-0.5 (~3B params), aiming for a single model that can control many different robot embodiments -- a direct competitor to systems like NVIDIA GR00T.

FILING BIS-F/2026-0479 · SYZYGY 6::3 · ZONE 3 · HORIZON 2028 · SWERVE 6.640

The Foundation Network publishes the first production-grade physical-intelligence agent: a trained physical intelligence model released as an open-source node module alongside its training artifacts, deployed into a controlled testbed where it performs tasks against multiple robots across a domain without retraining. The model's performance curve is logged publicly along with every hardware variant tested. Resolution obtains when the Foundation Network Node Model Registry releases the published agent version number under this filing, dated Q4 2028.

RESOLUTION SOURCE: Foundation Network Node Model Registry release notes · STATUS: PENDING

Skild AI

Builds a general-purpose 'omni-bodied' robot foundation model — a single shared AI brain trained to operate many different robot embodiments across tasks.

FILING BIS-F/2026-0372 · SYZYGY 5::4 · ZONE 4 · HORIZON 2029 · SWERVE 7.100

A robotic manufacturing line at a major carmaker logs a shift change where every workstation picks up a part it had previously passed by — no human picking hands are observed entering the cell — after running the Skild omni-body agent in place for eight weeks. The manufacturer's published productivity gain does not name the cause as Skild. The swerve is small and quiet: a machine learning that can drive itself through the plant becomes the implicit hand behind the conveyor belt. Resolution via SEMI Supply Chain Insights and the automaker's quarterly production disclosure.

RESOLUTION SOURCE: SEMI Supply Chain Insights; company Q4 production disclosure · STATUS: PENDING

ENERGY

Koloma

Denver-based geologic ('natural'/white) hydrogen company exploring for and aiming to produce naturally occurring subsurface hydrogen; backed by Breakthrough Energy Ventures, Khosla, Amazon Climate Pledge Fund (~$400M raised).

FILING BIS-F/2026-0735 · SYZYGY 6::3 · ZONE 3 · HORIZON 2030 · SWERVE 5.682

Koloma's first commercial hydrogen production from natural gas — not electrolysis, which it files as a demonstration — arrives at the Boulder County landfill under the state's mandatory diversion rule, fed into a composting loop that converts methane to electricity. The swerve is small enough to be counted in units rather than words: one million cubic meters diverted annually from combustion toward fertilizer. Resolution via Colorado Division of Air Quality permits filings and county sustainability reporting.

RESOLUTION SOURCE: Colorado Division of Air Quality permit filings + county sustainability reporting · STATUS: PENDING

Terraform Industries

Develops modular systems that synthesize pipeline-grade natural gas (methane) from atmospheric CO2, water, and cheap solar electricity; founder/CEO Casey Handmer.

FILING BIS-F / DEVELOPMENT-ENVIRONMENT. A Texas industrial park will host the first commercial-scale demonstration of Terraform's fully synthetic methane plant — flue-cured, low-carbon, no fossil fuel-fired generation in sight. The operator files commissioning milestones under ERCOT's transmission interconnection permitting process. Swerve: the first megawatt-hour produced from nothing more than sunlight, air, and water arrives at the grid, and every downstream burner must learn how to burn it. Resolution via ERCOT interconnection permit filings and the company's SEC disclosure schedule.

RESOLUTION SOURCE: ERCOT interconnection permit filings + Terraform Industries SEC disclosure schedule (Securities and Exchange Commission)

STATUS: PENDING

ARTIFICIAL INTELLIGENCE

Anthropic

San Francisco AI safety and research company; builds the Claude LLM family (latest Claude Opus 4.8) and the Claude Code developer product, reporting a ~$47B revenue run rate.

FILING BIS-F/2026-0330 · SYZYGY 5::4 · ZONE 5 · HORIZON 2033 · SWERVE 6.144

The Bureau records that before the end of calendar year 2033, Anthropic ships a Mythos-class model whose inference pipeline trains itself to select its own parameterization from the prior distribution — no human expert choosing anything, the model learning which parameters it can trust to be stable without being told they're safe, and selecting them as such, in full autonomy. The post-release disclosure will log the self-training regime as deliberate. This swerve is small enough to reverse: if Anthropic chooses to keep the selection opaque, the filing collapses into the neutral claim every model makes. Resolution via the published training-metadata paper under arXiv.org.

RESOLUTION SOURCE: arXiv.org preprint server (training-metadata paper); Anthropic blog ·
STATUS: PENDING

OpenAI

Frontier AI research company behind ChatGPT and the GPT model family (latest GPT-5.5) plus the Codex agentic-coding platform; weighing an IPO around 2027 at a ~$1T target.

FILING BIS-F/2026-0443 · SYZYGY 6::3 · ZONE 6 · HORIZON 2027 · SWERVE 5.414

OpenAI files that its annual revenue will exceed $1 billion before year-end 2027; it publishes the quarterly results confirming this. The filing names no event outside the report itself and resolves against the publicly filed Q4 2026 and 2027 quarters' revenues disclosed under SEC disclosure rules. It does not predict whether anyone bought anything from OpenAI's products; it simply states whether they have crossed their threshold.

RESOLUTION SOURCE: SEC Form 10-Q filings — Revenue line item; operator disclosures · STATUS: PENDING

BIOTECH

Colossal Biosciences

Dallas-based de-extinction and species-preservation company using genome engineering to revive extinct animals (woolly mammoth, dire wolf, dodo, moa, thylacine, bluebuck) and build spinoff conservation technologies.

FILING BIS-F/2026-0574 · SYZYGY 5::4 · ZONE 5 · HORIZON 2027 · SWERVE 6.218

Colossal's Bluebuck Conservation Technologies unit reports the first artificial embryo hatch at a laboratory outside Botswana, demonstrating that a critically endangered Afrotropical antelope can be bred anew without native females by fertilising eggs harvested from frozen-thawed sperm stock. The result is published in an open-access journal and filed under FAO/CABI guidelines for captive-breeding programmes. The swerve is small — one clutch, one generation — yet it closes a decades-long gap between extinction risk and recovery strategy. Resolution via CABIAI/FAO captive-breeding registry.

RESOLUTION SOURCE: CABIAI/Fao captive-breeding registry / arXiv preprint (Bluebuck Conservation Technologies report) · STATUS: PENDING

Retro Biosciences

Sam Altman-backed longevity biotech pursuing cellular reprogramming plus autophagy and plasma-based therapies to extend human healthspan; valued ~$1.8B in 2026.

FILING BIS-F/2730 · SYZYGY 5::4 · ZONE 5 · HORIZON 2030 · SWERVE 6.740

The first clinical trial at a US site reports that allogeneic hematopoietic stem cells harvested from donors under twenty years old and treated by Retro's cell-reprogramming platform produce functional adult tissue within two months of transplantation, meeting regulatory endpoint criteria without requiring bone marrow expansion. The patient registry logs the transplant; the FDA investigational-new-drug letter publishes the endpoints. The

swerve is modest — no rejuvenated organs delivered yet — but it is also the point where a therapy gains meaningful applicability across age groups.

RESOLUTION SOURCE: ClinicalTrials.gov registration + FDA IND / PDUFA guidance document · STATUS: PENDING